System Modelling Techniques - I

Functions of Several Variables, Partial Derivatives, Chain Rule, Implicit Functions, Exact Differentials, and Potential Function Reconstruction

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1 Functions of Several Variables, Limits, and Continuity

Definition 1.1 (Scalar Field / Function of Several Variables)**.** Let $D \subseteq \mathbb{R}^n$ be a non-empty domain. A *real-valued function of n variables* f is a mapping that assigns to every vector $\mathbf{x} = (x_1, x_2, \ldots, x_n) \in D$ a unique real number $z = f(\mathbf{x}) \in \mathbb{R}$. We write $f : D \to \mathbb{R}$.

Definition 1.2 (Open Ball and Topological Concepts)**.** Let $\mathbf{x}_0 \in \mathbb{R}^n$ and $r > 0$.

1. The *open ball* centered at $\mathbf{x}_0$ with radius r is:
$$B_r(\mathbf{x}_0) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| < r\} \tag{1}$$
where $\|\mathbf{x}\| = \sqrt{\sum_{i=1}^{n} x_i^2}$ is the standard Euclidean norm.
2. A point $\mathbf{x}_0$ is an *interior point* of D if there exists $r > 0$ such that $B_r(\mathbf{x}_0) \subseteq D$.
3. A point $\mathbf{x}_0$ is a *limit point* of D if every open ball $B_r(\mathbf{x}_0)$ contains at least one point of D distinct from $\mathbf{x}_0$.

Definition 1.3 (Multivariable Limit)**.** Let $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ and let $\mathbf{x}_0$ be a limit point of D. The *limit of* $f(\mathbf{x})$ *as* $\mathbf{x}$ *approaches* $\mathbf{x}_0$ *is* L, written:
$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = L \tag{2}$$
if for every $\epsilon > 0$, there exists a corresponding $\delta > 0$ such that for all $\mathbf{x} \in D$:
$$0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta \implies |f(\mathbf{x}) - L| < \epsilon \tag{3}$$

Theorem 1.1 (Path-Dependence Test for Limit Non-Existence)**.** *If* $f(\mathbf{x})$ *approaches two different values along two distinct continuous paths approaching* $\mathbf{x}_0$*, then* $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x})$ *does not exist.*

Proof. Suppose $\gamma_1(t)$ and $\gamma_2(t)$ are two continuous paths in $D \backslash \{\mathbf{x}_0\}$ such that $\lim_{t\to 0} \gamma_1(t) = \mathbf{x}_0$ and $\lim_{t\to 0} \gamma_2(t) = \mathbf{x}_0$. Assume $\lim_{t\to 0} f(\gamma_1(t)) = L_1$ and $\lim_{t\to 0} f(\gamma_2(t)) = L_2$ with $L_1 \neq L_2$. Choose $\epsilon = \frac{|L_1 - L_2|}{2} > 0$. If $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = L$, then by definition $f(\mathbf{x})$ must enter $B_\epsilon(L)$ for all $\mathbf{x}$ sufficiently close to $\mathbf{x}_0$, which forces $L_1 = L_2 = L$, contradicting $L_1 \neq L_2$. □

Definition 1.4 (Continuity)**.** A function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is *continuous at* $\mathbf{x}_0 \in D$ if $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$.

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2 Partial Derivatives and Total Differentiability

2.1 Definitions and Fundamental Relations

Definition 2.1 (Partial Derivative)**.** Let $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ and $\mathbf{x}_0 = (x_{1,0}, \ldots, x_{n,0}) \in \text{int}(D)$. The *partial derivative* of f with respect to x_i at $\mathbf{x}_0$ is:
$$\frac{\partial f}{\partial x_i}(\mathbf{x}_0) = \lim_{h\to 0} \frac{f(x_{1,0}, \ldots, x_{i,0} + h, \ldots, x_{n,0}) - f(x_{1,0}, \ldots, x_{i,0}, \ldots, x_{n,0})}{h} \tag{4}$$

Definition 2.2 (Total Differentiability)**.** A function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is *differentiable* at an interior point $\mathbf{x}_0 \in D$ if there exists a linear transformation represented by a vector $\mathbf{A} \in \mathbb{R}^n$ such that:

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{f(\mathbf{x}_0+\mathbf{h}) - f(\mathbf{x}_0) - \mathbf{A}\cdot\mathbf{h}}{\|\mathbf{h}\|} = 0 \tag{5}$$

When such $\mathbf{A}$ exists, $\mathbf{A} = \nabla f(\mathbf{x}_0) = \left(\frac{\partial f}{\partial x_1}(\mathbf{x}_0), \ldots, \frac{\partial f}{\partial x_n}(\mathbf{x}_0)\right)$, and the total differential is defined as $df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} dx_i$.

2.2 Key Structural Theorems

Theorem 2.1 (Differentiability Implies Continuity)**.** *If $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}_0 \in D$, then f is continuous at $\mathbf{x}_0$.*

Proof. Let $\mathbf{h} = \mathbf{x} - \mathbf{x}_0$. By differentiability:

$$f(\mathbf{x}_0+\mathbf{h}) - f(\mathbf{x}_0) = \nabla f(\mathbf{x}_0)\cdot\mathbf{h} + E(\mathbf{h}) \tag{6}$$

where $\lim_{\mathbf{h}\to\mathbf{0}} \frac{E(\mathbf{h})}{\|\mathbf{h}\|} = 0$. Taking limits as $\mathbf{h} \to \mathbf{0}$:

$$\begin{aligned}\lim_{\mathbf{h}\to\mathbf{0}} |f(\mathbf{x}_0+\mathbf{h}) - f(\mathbf{x}_0)| &\le \lim_{\mathbf{h}\to\mathbf{0}} \|\nabla f(\mathbf{x}_0)\|\|\mathbf{h}\| + \lim_{\mathbf{h}\to\mathbf{0}} \left(\frac{|E(\mathbf{h})|}{\|\mathbf{h}\|}\|\mathbf{h}\|\right) \\ &= \|\nabla f(\mathbf{x}_0)\|\cdot 0 + 0\cdot 0 = 0 \end{aligned} \tag{7}$$

Thus $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$, proving continuity. □

Theorem 2.2 (Sufficient Condition for Differentiability)**.** *Let $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$. If the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist in an open ball $B_r(x_0, y_0)$ and are continuous at (x_0, y_0), then f is differentiable at (x_0, y_0).*

Proof. Let $\mathbf{h} = (h, k)$ such that $(x_0 + h, y_0 + k) \in B_r(x_0, y_0)$. Decompose the change in f:

$$\Delta f = f(x_0+h, y_0+k) - f(x_0, y_0) = [f(x_0+h, y_0+k) - f(x_0, y_0+k)] + [f(x_0, y_0+k) - f(x_0, y_0)] \tag{8}$$

Applying the 1D Mean Value Theorem to each bracketed single-variable slice:

$$f(x_0+h, y_0+k) - f(x_0, y_0+k) = h\frac{\partial f}{\partial x}(x_0+\theta_1 h, y_0+k), \quad \theta_1 \in (0,1) \tag{9}$$

$$f(x_0, y_0+k) - f(x_0, y_0) = k\frac{\partial f}{\partial y}(x_0, y_0+\theta_2 k), \quad \theta_2 \in (0,1) \tag{10}$$

Define error terms $\epsilon_1(h,k) = \frac{\partial f}{\partial x}(x_0+\theta_1 h, y_0+k) - \frac{\partial f}{\partial x}(x_0, y_0)$ and $\epsilon_2(h,k) = \frac{\partial f}{\partial y}(x_0, y_0+\theta_2 k) - \frac{\partial f}{\partial y}(x_0, y_0)$. Continuous partials ensure $\epsilon_1, \epsilon_2 \to 0$ as $(h,k) \to (0,0)$. Then:

$$\Delta f = h\frac{\partial f}{\partial x}(x_0, y_0) + k\frac{\partial f}{\partial y}(x_0, y_0) + h\epsilon_1 + k\epsilon_2 \tag{11}$$

Dividing by $\sqrt{h^2+k^2}$:

$$\frac{|h\epsilon_1 + k\epsilon_2|}{\sqrt{h^2+k^2}} \le \frac{|h|}{\sqrt{h^2+k^2}}|\epsilon_1| + \frac{|k|}{\sqrt{h^2+k^2}}|\epsilon_2| \le |\epsilon_1| + |\epsilon_2| \xrightarrow{(h,k)\to(0,0)} 0 \tag{12}$$

Hence, f is totally differentiable at (x_0, y_0). □

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Theorem 2.3 (Clairaut's / Schwarz's Theorem on Mixed Partials)**.** *Let $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$. If f_{xy} and f_{yx} exist and are continuous on an open disk centered at (x_0, y_0), then $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$.*

Proof. Consider $\Delta(h,k) = f(x_0+h, y_0+k) - f(x_0+h, y_0) - f(x_0, y_0+k) + f(x_0, y_0)$. Defining $\phi(x) = f(x, y_0+k) - f(x, y_0)$, we have $\Delta(h,k) = \phi(x_0+h) - \phi(x_0)$. Applying MVT twice:

$$\Delta(h,k) = h\phi'(x_0+\theta_1 h) = hk f_{xy}(x_0+\theta_1 h, y_0+\theta_2 k), \quad \theta_1, \theta_2 \in (0,1) \tag{13}$$

Defining $\psi(y) = f(x_0+h, y) - f(x_0, y)$ and applying MVT twice in reverse order:

$$\Delta(h,k) = \psi(y_0+k) - \psi(y_0) = hk f_{yx}(x_0+\theta_3 h, y_0+\theta_4 k), \quad \theta_3, \theta_4 \in (0,1) \tag{14}$$

Equating both expressions, dividing by $hk \neq 0$, and taking $(h,k) \to (0,0)$ via continuity yields $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$. □

3 The Multivariable Chain Rule

Theorem 3.1 (Chain Rule for Single Parameter)**.** *Let $x = x(t), y = y(t)$ be differentiable at t_0, and $z = f(x,y)$ be differentiable at $(x(t_0), y(t_0))$. Then:*

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} \tag{15}$$

Proof. Let $\Delta x = x(t_0+\Delta t) - x(t_0)$, $\Delta y = y(t_0+\Delta t) - y(t_0)$. Differentiability of f implies:

$$\Delta z = \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y \tag{16}$$

where $\epsilon_1, \epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$. Dividing by $\Delta t \neq 0$ and taking $\Delta t \to 0$:

$$\frac{dz}{dt} = \lim_{\Delta t \to 0}\frac{\Delta z}{\Delta t} = \frac{\partial f}{\partial x}\lim_{\Delta t \to 0}\frac{\Delta x}{\Delta t} + \frac{\partial f}{\partial y}\lim_{\Delta t \to 0}\frac{\Delta y}{\Delta t} + 0 \cdot \frac{dx}{dt} + 0 \cdot \frac{dy}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} \tag{17}$$

□

Theorem 3.2 (Matrix Formulation of General Chain Rule)**.** *Let $\mathbf{g} : \mathbb{R}^m \to \mathbb{R}^n$ be differentiable at $\mathbf{u}_0$, and $f : \mathbb{R}^n \to \mathbb{R}$ differentiable at $\mathbf{x}_0 = \mathbf{g}(\mathbf{u}_0)$. The Jacobian of $h = f \circ \mathbf{g}$ satisfies:*

$$J_h(\mathbf{u}_0) = J_f(\mathbf{g}(\mathbf{u}_0)) \cdot J_{\mathbf{g}}(\mathbf{u}_0) \iff \frac{\partial z}{\partial u_j} = \sum_{i=1}^{n} \frac{\partial z}{\partial x_i}\frac{\partial x_i}{\partial u_j} \tag{18}$$

4 Differentiation of Implicit Functions

Theorem 4.1 (Implicit Function Theorem for Single Relation $F(x,y) = 0$)**.** *Let $F : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ be C^1. If $F(x_0, y_0) = 0$ and $F_y(x_0, y_0) \neq 0$, there exists an open interval I containing x_0 and a unique C^1 function $y = g(x)$ on I such that $F(x, g(x)) = 0$ and:*

$$\frac{dy}{dx} = -\frac{F_x(x,y)}{F_y(x,y)} \tag{19}$$

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Proof. Differentiating $F(x, y(x)) = 0$ with respect to x using the chain rule:

$$\frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0 \implies F_x + F_y\frac{dy}{dx} = 0 \tag{20}$$

Since $F_y \neq 0$ near (x_0, y_0), solving for $\frac{dy}{dx}$ gives $\frac{dy}{dx} = -\frac{F_x}{F_y}$. □

Theorem 4.2 (Implicit System Theorem / Jacobian Formulation)**.** *Let $F(x, y, u, v) = 0$ and $G(x, y, u, v) = 0$. If the Jacobian determinant of (F, G) with respect to (u, v) is non-zero:*

$$\frac{\partial(F, G)}{\partial(u, v)} = \begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix} \neq 0 \tag{21}$$

then u and v are differentiable functions of x and y, and:

$$\frac{\partial u}{\partial x} = -\frac{\frac{\partial(F,G)}{\partial(x,v)}}{\frac{\partial(F,G)}{\partial(u,v)}}, \quad \frac{\partial v}{\partial x} = -\frac{\frac{\partial(F,G)}{\partial(u,x)}}{\frac{\partial(F,G)}{\partial(u,v)}} \tag{22}$$

Proof. Differentiate $F = 0$ and $G = 0$ partially with respect to x:

$$F_x + F_u\frac{\partial u}{\partial x} + F_v\frac{\partial v}{\partial x} = 0 \tag{23}$$

$$G_x + G_u\frac{\partial u}{\partial x} + G_v\frac{\partial v}{\partial x} = 0 \tag{24}$$

In matrix form:

$$\begin{bmatrix} F_u & F_v \\ G_u & G_v \end{bmatrix}\begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial x} \end{bmatrix} = -\begin{bmatrix} F_x \\ G_x \end{bmatrix} \tag{25}$$

Applying Cramer's Rule yields the desired expressions. □

5 Exact Differentials, Integrating Factors, and Reconstructing the Potential Function

5.1 Definition and Exactness Conditions

Definition 5.1 (Exact Differential Form)**.** The differential form $\omega = M(x, y)\,dx + N(x, y)\,dy$ is *exact* on a domain $D \subseteq \mathbb{R}^2$ if there exists a scalar function $f(x, y)$ (called the *potential function*) such that $df = \omega$, meaning:

$$\frac{\partial f}{\partial x} = M(x, y) \quad \text{and} \quad \frac{\partial f}{\partial y} = N(x, y) \tag{26}$$

In three variables, $\omega = M\,dx + N\,dy + P\,dz$ is exact if $df = M\,dx + N\,dy + P\,dz$, which corresponds to $\nabla f = (M, N, P)$.

Theorem 5.1 (Poincaré Criterion for Exactness)**.** *Let M, N, P be C^1 functions on a simply connected domain D.*

1. *$M\,dx + N\,dy$ is exact $\iff \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.*
2. *$M\,dx + N\,dy + P\,dz$ is exact $\iff$ curl$(M, N, P) = \mathbf{0}$, i.e.:*

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}, \quad \frac{\partial P}{\partial x} = \frac{\partial M}{\partial z} \tag{27}$$

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5.2 How to Answer: "What is the function whose differential is given?"

When faced with the fundamental problem:

"Given the differential form $df = M(x,y)dx + N(x,y)dy$, find the function $f(x,y)$,"

three complementary systematic methods can be employed.

5.2.1 Method 1: Partial Integration Algorithm (Indefinite Integration Method)

This is the standard procedural approach for both 2D and 3D forms.

Algorithm 5.1 (2D Potential Reconstruction Protocol)**.**

1. **Verify Exactness:** Compute $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$. Confirm $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
2. **Partial Integration w.r.t. x:** Integrate $M(x,y)$ with respect to x, treating y as a constant:
$$f(x,y) = \int M(x,y)\,dx + g(y) \tag{28}$$
where $g(y)$ is an unknown constant of integration with respect to x.
3. **Differentiate w.r.t. y:** Compute the partial derivative of the expression from Step 2:
$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(\int M(x,y)\,dx\right) + g'(y) \tag{29}$$
4. **Match with $N(x,y)$ and Solve for $g(y)$:** Set $\frac{\partial f}{\partial y} = N(x,y)$:
$$N(x,y) = \frac{\partial}{\partial y}\left(\int M(x,y)\,dx\right) + g'(y) \implies g'(y) = N(x,y) - \frac{\partial}{\partial y}\left(\int M(x,y)\,dx\right) \tag{30}$$
(Due to exactness, all x-dependent terms in $g'(y)$ will cancel out perfectly.)
5. **Integrate $g'(y)$:** Calculate $g(y) = \int g'(y)\,dy + C$.
6. **State Final Solution:** Substitute $g(y)$ back into Step 2:
$$f(x,y) = \int M(x,y)\,dx + g(y) + C \tag{31}$$

Algorithm 5.2 (3D Potential Reconstruction Protocol ($df = Mdx + Ndy + Pdz$))**.**

1. Integrate M w.r.t. x: $f(x,y,z) = \int M\,dx + g(y,z)$.
2. Differentiate w.r.t. y and set equal to N: $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(\int M\,dx\right) + \frac{\partial g}{\partial y} = N \implies \frac{\partial g}{\partial y} = H(y,z)$.
3. Integrate $\frac{\partial g}{\partial y}$ w.r.t. y: $g(y,z) = \int H(y,z)\,dy + h(z)$.
4. Differentiate $f(x,y,z) = \int M\,dx + \int H\,dy + h(z)$ w.r.t. z and match with P:
$$\frac{\partial f}{\partial z} = P(x,y,z) \implies h'(z) = K(z) \implies h(z) = \int K(z)\,dz + C \tag{32}$$

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5.2.2 Method 2: Definite Line Integral / Path Independence Method

Since $\omega = df$ is exact, line integrals $\int_C \omega$ depend only on the endpoints. We can reconstruct $f(x, y)$ by integrating along a convenient piecewise linear path from a reference basepoint (x_0, y_0) to a general point (x, y).

Theorem 5.2 (Definite Path Integration Formula)**.** *If $M\,dx + N\,dy$ is exact on D, choosing a reference point $(x_0, y_0) \in D$, the function $f(x, y)$ is given directly by:*

$$f(x, y) = \int_{x_0}^{x} M(t, y_0)\, dt + \int_{y_0}^{y} N(x, t)\, dt + C \tag{33}$$

Proof. Consider the two-segment path from $(x_0, y_0) \to (x, y_0) \to (x, y)$:

- **Segment 1:** (t, y_0) for $t \in [x_0, x]$, where $dy = 0$. Integral is $\int_{x_0}^{x} M(t, y_0)dt$.
- **Segment 2:** (x, t) for $t \in [y_0, y]$, where $dx = 0$. Integral is $\int_{y_0}^{y} N(x, t)dt$.

Summing these gives the result. □

5.2.3 Method 3: Grouping / Differential Inspection Method

For rapid solving, one can recognize fundamental combination building blocks.

Table 1: Standard Building Blocks for Differential Inspection

Differential Form Pattern	**Exact Equivalent** $d(f)$
$x\,dy + y\,dx$	$d(xy)$
$\frac{y\,dx - x\,dy}{y^2}$	$d\left(\frac{x}{y}\right)$
$\frac{x\,dy - y\,dx}{x^2}$	$d\left(\frac{y}{x}\right)$
$\frac{x\,dx + y\,dy}{x^2+y^2}$	$d\left(\frac{1}{2}\ln(x^2+y^2)\right)$
$\frac{x\,dy - y\,dx}{x^2+y^2}$	$d\left(\arctan\left(\frac{y}{x}\right)\right)$
$\frac{x\,dy - y\,dx}{xy}$	$d\left(\ln\left(\frac{y}{x}\right)\right)$

6 Fully Solved Practice Problems

Example 6.1 (Complete Potential Reconstruction - Trigonometric/Polynomial)**.** Find the function $f(x, y)$ whose differential is given by:

$$df = (2x\cos y - y^2\sin x)\,dx + (-x^2\sin y + 2y\cos x)\,dy \quad \text{with } f(0, 0) = 5 \tag{34}$$

Solution. **Step 1: Identify components and test exactness.**

$$M(x, y) = 2x\cos y - y^2\sin x, \quad N(x, y) = -x^2\sin y + 2y\cos x \tag{35}$$

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$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(2x\cos y - y^2\sin x) = -2x\sin y - 2y\sin x \tag{36}$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(-x^2\sin y + 2y\cos x) = -2x\sin y - 2y\sin x \tag{37}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, an exact potential $f(x,y)$ exists.

Step 2: Apply Partial Integration w.r.t. x.

$$f(x,y) = \int M(x,y)\,dx + g(y) = \int (2x\cos y - y^2\sin x)\,dx + g(y) \tag{38}$$

$$f(x,y) = x^2\cos y + y^2\cos x + g(y) \tag{39}$$

Step 3: Differentiate w.r.t. y and equate to $N(x,y)$.

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2\cos y + y^2\cos x + g(y)) = -x^2\sin y + 2y\cos x + g'(y) \tag{40}$$

Set equal to $N(x,y)$:

$$-x^2\sin y + 2y\cos x + g'(y) = -x^2\sin y + 2y\cos x \implies g'(y) = 0 \tag{41}$$

Step 4: Solve for $g(y)$ and apply initial condition. Integration gives $g(y) = C$. Thus:

$$f(x,y) = x^2\cos y + y^2\cos x + C \tag{42}$$

Apply $f(0,0) = 5$:

$$f(0,0) = 0^2\cos(0) + 0^2\cos(0) + C = 5 \implies C = 5 \tag{43}$$

Final Result:

$$f(x,y) = x^2\cos y + y^2\cos x + 5 \tag{44}$$

□

Example 6.2 (3D Potential Function Reconstruction)**.** Find the scalar function $f(x,y,z)$ whose differential is:

$$df = (2xy + z^3)\,dx + (x^2 + 3y^2z)\,dy + (3xz^2 + y^3)\,dz \tag{45}$$

Solution. **Step 1: Check Exactness in 3D.** $M = 2xy + z^3, N = x^2 + 3y^2z, P = 3xz^2 + y^3$.

$$M_y = 2x, \quad N_x = 2x \implies M_y = N_x \tag{46}$$

$$N_z = 3y^2, \quad P_y = 3y^2 \implies N_z = P_y \tag{47}$$

$$P_x = 3z^2, \quad M_z = 3z^2 \implies P_x = M_z \tag{48}$$

The field is conservative and df is exact.

Step 2: Sequential Integration Method.

1. $f(x,y,z) = \int (2xy + z^3)\,dx = x^2y + xz^3 + g(y,z)$.
2. $\frac{\partial f}{\partial y} = x^2 + \frac{\partial g}{\partial y} = N = x^2 + 3y^2z \implies \frac{\partial g}{\partial y} = 3y^2z \implies g(y,z) = y^3z + h(z)$.
3. $f(x,y,z) = x^2y + xz^3 + y^3z + h(z)$.

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4. $\frac{\partial f}{\partial z} = 3xz^2 + y^3 + h'(z) = P = 3xz^2 + y^3 \implies h'(z) = 0 \implies h(z) = C.$

Final Result:

$$f(x, y, z) = x^2y + xz^3 + y^3z + C \tag{49}$$

□

Example 6.3 (Implicit Surface Partial Derivative Evaluation)**.** Given $x^3+y^3+z^3+6xyz = 1$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(1, 1, -1)$ using the Implicit Function Theorem.

Solution. Let $F(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1 = 0$.

Compute partial derivatives of F:

$$F_x = 3x^2 + 6yz \tag{50}$$
$$F_y = 3y^2 + 6xz \tag{51}$$
$$F_z = 3z^2 + 6xy \tag{52}$$

Evaluate partial derivatives at $P(1, 1, -1)$:

$$F_x(1, 1, -1) = 3(1)^2 + 6(1)(-1) = 3 - 6 = -3 \tag{53}$$
$$F_y(1, 1, -1) = 3(1)^2 + 6(1)(-1) = 3 - 6 = -3 \tag{54}$$
$$F_z(1, 1, -1) = 3(-1)^2 + 6(1)(1) = 3 + 6 = 9 \tag{55}$$

Since $F_z(1, 1, -1) = 9 \neq 0$, z is implicitly differentiable near P. Apply the Implicit Derivative Corollary:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{-3}{9} = \frac{1}{3} \tag{56}$$
$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{-3}{9} = \frac{1}{3} \tag{57}$$

□

7 Integrating Factors: Transforming Inexact Differentials to Exact Differentials

When a differential form $\omega = M(x, y)\,dx + N(x, y)\,dy$ is *inexact* ($\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$), it cannot be written directly as df for any scalar potential f. However, under general smoothness conditions, there exists a multiplier function $\mu(x, y) \neq 0$ that forces the product form $\mu\omega$ to become exact.

Definition 7.1 (Integrating Factor)**.** A non-vanishing function $\mu : D \subseteq \mathbb{R}^2 \to \mathbb{R} \setminus \{0\}$ is called an *integrating factor* for the differential form $\omega = M(x, y)\,dx + N(x, y)\,dy$ on D if the modified form:

$$\omega_\mu = \mu(x, y)M(x, y)\,dx + \mu(x, y)N(x, y)\,dy \tag{58}$$

is exact on D. That is, there exists a continuously differentiable function $f(x, y)$ such that $df = \mu M\,dx + \mu N\,dy$.

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7.1 The Fundamental Governing PDE

Theorem 7.1 (Governing Partial Differential Equation for Integrating Factors)**.** *Let $M, N \in C^1(D)$ and $\mu \in C^1(D)$ with $\mu(x,y) \neq 0$. The function $\mu(x,y)$ is an integrating factor for $M\,dx + N\,dy$ if and only if μ satisfies the linear first-order partial differential equation:*

$$N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)\mu \tag{59}$$

Proof. By Poincaré's Lemma, $\mu M\,dx + \mu N\,dy$ is exact on a simply connected domain D if and only if:

$$\frac{\partial}{\partial y}\left[\mu(x,y)M(x,y)\right] = \frac{\partial}{\partial x}\left[\mu(x,y)N(x,y)\right] \tag{60}$$

Applying the product rule for partial derivatives to both sides:

$$\frac{\partial \mu}{\partial y}M + \mu\frac{\partial M}{\partial y} = \frac{\partial \mu}{\partial x}N + \mu\frac{\partial N}{\partial x} \tag{61}$$

Rearranging terms with μ on the right and its partial derivatives on the left:

$$N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)\mu \tag{62}$$

□

Remark 7.1. Solving the general PDE $N\mu_x - M\mu_y = (M_y - N_x)\mu$ for an arbitrary two-variable function $\mu(x,y)$ is generally as difficult as solving the original differential equation. Therefore, we search for structural reduction criteria where μ depends only on a single composite variable $z = z(x,y)$.

7.2 Single-Variable Integrating Factor Criteria

Lemma 7.2 (Integrating Factor Depending Only on x)**.** *If the quantity:*

$$f(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \tag{63}$$

is a function of x alone (independent of y), then $\mu = \mu(x)$ is a single-variable integrating factor given by:

$$\mu(x) = \exp\left(\int f(x)\,dx\right) \tag{64}$$

Proof. Assume μ depends strictly on x, so $\frac{\partial \mu}{\partial x} = \frac{d\mu}{dx}$ and $\frac{\partial \mu}{\partial y} = 0$. Substitute these into the governing PDE from Theorem 7.1:

$$N\frac{d\mu}{dx} - M(0) = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)\mu \implies \frac{1}{\mu}\frac{d\mu}{dx} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \tag{65}$$

Since the right-hand side is $f(x)$, we have an ordinary separable differential equation:

$$\frac{d}{dx}\left(\ln|\mu|\right) = f(x) \implies \ln|\mu(x)| = \int f(x)\,dx \implies \mu(x) = \exp\left(\int f(x)\,dx\right) \tag{66}$$

□

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Lemma 7.3 (Integrating Factor Depending Only on y)**.** *If the quantity:*

$$g(y) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} \tag{67}$$

is a function of y alone (independent of x), then $\mu = \mu(y)$ is a single-variable integrating factor given by:

$$\mu(y) = \exp\left(\int g(y)\, dy\right) \tag{68}$$

Proof. Assume μ depends strictly on y, so $\frac{\partial \mu}{\partial x} = 0$ and $\frac{\partial \mu}{\partial y} = \frac{d\mu}{dy}$. Substituting into Theorem 7.1:

$$N(0) - M\frac{d\mu}{dy} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)\mu \implies -M\frac{d\mu}{dy} = -\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\mu \tag{69}$$

Dividing by $-M\mu$:

$$\frac{1}{\mu}\frac{d\mu}{dy} = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = g(y) \implies \mu(y) = \exp\left(\int g(y)\, dy\right) \tag{70}$$

□

7.3 General Composite Variable Integrating Factors

Theorem 7.4 (Composite Variable Criterion $\mu = \mu(z(x,y))$)**.** *Let $z = z(x,y)$ be a given continuously differentiable composite function. If the expression:*

$$h(z) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N\frac{\partial z}{\partial x} - M\frac{\partial z}{\partial y}} \tag{71}$$

can be expressed purely as a function of z, then $\mu(z) = \exp\left(\int h(z)\, dz\right)$ is an integrating factor.

Proof. Let $\mu(x,y) = \mu(z(x,y))$. By the multivariable chain rule:

$$\frac{\partial \mu}{\partial x} = \frac{d\mu}{dz}\frac{\partial z}{\partial x}, \quad \frac{\partial \mu}{\partial y} = \frac{d\mu}{dz}\frac{\partial z}{\partial y} \tag{72}$$

Substitute these chain rule expansions into Theorem 7.1:

$$N\left(\frac{d\mu}{dz}\frac{\partial z}{\partial x}\right) - M\left(\frac{d\mu}{dz}\frac{\partial z}{\partial y}\right) = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)\mu \tag{73}$$

Factor out $\frac{d\mu}{dz}$:

$$\frac{d\mu}{dz}\left(N\frac{\partial z}{\partial x} - M\frac{\partial z}{\partial y}\right) = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)\mu \tag{74}$$

Dividing both sides by $\mu\left(N\frac{\partial z}{\partial x} - M\frac{\partial z}{\partial y}\right)$:

$$\frac{1}{\mu}\frac{d\mu}{dz} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N\frac{\partial z}{\partial x} - M\frac{\partial z}{\partial y}} = h(z) \tag{75}$$

Integrating with respect to z yields $\mu(z) = \exp\left(\int h(z)\, dz\right)$. □

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Corollary 7.5 (Special Composite Variables)**.**

1. ***Product Variable*** *($z = xy$)**:*** *If $\frac{M_y - N_x}{yN - xM} = h(xy)$, then $\mu(xy) = \exp\left(\int h(z)\,dz\right)$.*
2. ***Sum of Squares*** *($z = x^2 + y^2$)**:*** *If $\frac{M_y - N_x}{2xN - 2yM} = h(x^2 + y^2)$, then $\mu(x^2 + y^2) = \exp\left(\int h(z)\,dz\right)$.*

7.4 Algebraic Criteria for Algebraic Structures

Theorem 7.6 (Euler's Theorem for Homogeneous Differential Forms)**.** *Let $M(x, y)$ and $N(x, y)$ be homogeneous functions of the same degree $n \in \mathbb{R}$. If $xM(x, y) + yN(x, y) \neq 0$, then:*

$$\mu(x, y) = \frac{1}{xM(x, y) + yN(x, y)} \tag{76}$$

is an integrating factor for $M\,dx + N\,dy$.

Proof. By Euler's Theorem on Homogeneous Functions, since M and N are homogeneous of degree n:

$$x\frac{\partial M}{\partial x} + y\frac{\partial M}{\partial y} = nM \quad \text{and} \quad x\frac{\partial N}{\partial x} + y\frac{\partial N}{\partial y} = nN \tag{77}$$

Define $D(x, y) = xM + yN$, so $\mu = \frac{1}{D}$. To prove exactness of $\mu M\,dx + \mu N\,dy$, we must show that $\frac{\partial}{\partial y}\left(\frac{M}{D}\right) = \frac{\partial}{\partial x}\left(\frac{N}{D}\right)$, which is equivalent to:

$$\frac{M_y D - M D_y}{D^2} = \frac{N_x D - N D_x}{D^2} \iff (M_y - N_x)D = MD_y - ND_x \tag{78}$$

Calculate partial derivatives of $D = xM + yN$:

$$D_x = M + xM_x + yN_x \tag{79}$$

$$D_y = xM_y + N + yN_y \tag{80}$$

Substitute D_x and D_y into the Right-Hand Side (RHS) of Equation (78):

$$\begin{aligned} \text{RHS} &= M(xM_y + N + yN_y) - N(M + xM_x + yN_x) \\ &= xMM_y + MN + yMN_y - MN - xNM_x - yNN_x \\ &= xMM_y + yMN_y - xNM_x - yNN_x \end{aligned} \tag{81}$$

Substitute $xM_x = nM - yM_y$ and $yN_y = nN - xN_x$ from Equation (77):

$$\begin{aligned} \text{RHS} &= xMM_y + M(nN - xN_x) - N(nM - yM_y) - yNN_x \\ &= xMM_y + nMN - xMN_x - nMN + yNM_y - yNN_x \\ &= xM(M_y - N_x) + yN(M_y - N_x) \\ &= (xM + yN)(M_y - N_x) = D(M_y - N_x) = \text{LHS} \end{aligned} \tag{82}$$

Since LHS = RHS, $\mu = \frac{1}{xM + yN}$ is an integrating factor. □

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7.5 Fully Solved Examples

Example 7.1 (Inexact Form Converted via $\mu(x)$ and Potential Recovery)**.** Check if $\omega = (3xy + y^2)\,dx + (x^2 + xy)\,dy$ is exact. If not, find an integrating factor $\mu(x)$, render the form exact, and reconstruct the scalar function $f(x, y)$ such that $df = \mu\omega$.

Solution. **1. Test for Exactness:** $M(x, y) = 3xy + y^2$ and $N(x, y) = x^2 + xy$.

$$\frac{\partial M}{\partial y} = 3x + 2y, \quad \frac{\partial N}{\partial x} = 2x + y \tag{83}$$

Since $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = (3x + 2y) - (2x + y) = x + y \neq 0$, the form is inexact.

2. Find Integrating Factor $\mu(x)$: Evaluate the ratio from Lemma 7.2:

$$f(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{x + y}{x^2 + xy} = \frac{x + y}{x(x + y)} = \frac{1}{x} \tag{84}$$

Since $f(x) = \frac{1}{x}$ is a function of x alone, an integrating factor is:

$$\mu(x) = \exp\left(\int \frac{1}{x}\,dx\right) = e^{\ln x} = x \tag{85}$$

3. Multiply and Confirm Exactness: Multiply ω by $\mu(x) = x$:

$$\omega_\mu = (3x^2y + xy^2)\,dx + (x^3 + x^2y)\,dy \implies M_{\text{new}} = 3x^2y + xy^2, \quad N_{\text{new}} = x^3 + x^2y \tag{86}$$

Verify exactness:

$$\frac{\partial M_{\text{new}}}{\partial y} = 3x^2 + 2xy, \quad \frac{\partial N_{\text{new}}}{\partial x} = 3x^2 + 2xy \quad \checkmark \tag{87}$$

4. Reconstruct Potential Function $f(x, y)$: Integrate M_{new} with respect to x:

$$f(x, y) = \int (3x^2y + xy^2)\,dx = x^3y + \frac{1}{2}x^2y^2 + g(y) \tag{88}$$

Differentiate w.r.t. y and match with N_{new}:

$$\frac{\partial f}{\partial y} = x^3 + x^2y + g'(y) = x^3 + x^2y \implies g'(y) = 0 \implies g(y) = C \tag{89}$$

Thus, the potential function whose differential is $x\omega$ is:

$$f(x, y) = x^3y + \frac{1}{2}x^2y^2 + C \tag{90}$$

□

Example 7.2 (Composite Variable Integrating Factor $\mu(x^2 + y^2)$)**.** Find an integrating factor depending on $z = x^2 + y^2$ for:

$$(x^3 + xy^2 - y)\,dx + (x^2y + y^3 + x)\,dy = 0 \tag{91}$$

and determine its corresponding potential function.

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Solution. **1. Test for Exactness:** $M = x(x^2+y^2) - y$ and $N = y(x^2+y^2) + x$.

$$M_y = 2xy - 1, \quad N_x = 2xy + 1 \implies M_y - N_x = -2 \neq 0 \tag{92}$$

2. Find Integrating Factor $\mu(z)$ where $z = x^2+y^2$: Apply Theorem 7.4 with $z_x = 2x$ and $z_y = 2y$:

$$h(z) = \frac{M_y - N_x}{Nz_x - Mz_y} = \frac{-2}{(x^2y + y^3 + x)(2x) - (x^3 + xy^2 - y)(2y)} \tag{93}$$

Expand denominator:

$$\begin{aligned} 2x(x^2y + y^3 + x) - 2y(x^3 + xy^2 - y) &= 2x^3y + 2xy^3 + 2x^2 - (2x^3y + 2xy^3 - 2y^2) \\ &= 2x^2 + 2y^2 = 2(x^2+y^2) = 2z \end{aligned} \tag{94}$$

Thus:

$$h(z) = \frac{-2}{2z} = -\frac{1}{z} \implies \mu(z) = \exp\left(\int -\frac{1}{z}\,dz\right) = \frac{1}{z} = \frac{1}{x^2+y^2} \tag{95}$$

3. Reconstruct Potential Function: Multiply by $\mu = \frac{1}{x^2+y^2}$:

$$\omega_\mu = \left(x - \frac{y}{x^2+y^2}\right)dx + \left(y + \frac{x}{x^2+y^2}\right)dy \tag{96}$$

Integrate $M_{\text{new}} = x - \frac{y}{x^2+y^2}$ w.r.t. x:

$$f(x,y) = \frac{x^2}{2} - \arctan\left(\frac{x}{y}\right) + g(y) \tag{97}$$

Differentiating w.r.t. y and setting equal to $N_{\text{new}} = y + \frac{x}{x^2+y^2}$ yields $g'(y) = y \implies g(y) = \frac{y^2}{2} + C$. Hence, the function is:

$$f(x,y) = \frac{x^2+y^2}{2} + \arctan\left(\frac{y}{x}\right) + C \tag{98}$$

□

8 Comprehensive Unsolved Practice Problems

1. **Potential Function Reconstruction (Polynomial-Exponential):** Find the function $f(x,y)$ whose differential is:
$$df = (ye^{xy} + 2x)\,dx + (xe^{xy} - 3y^2)\,dy, \quad f(0,0) = 2$$
2. **Potential Function Reconstruction (Logarithmic-Rational):** Reconstruct $f(x,y)$ from:
$$df = \left(\frac{1}{y} - \frac{y}{x^2}\right)dx + \left(\frac{1}{x} - \frac{x}{y^2}\right)dy$$
3. **3D Differential Form Function Recovery:** Determine the function $f(x,y,z)$ whose differential is:
$$df = (y^2z + 2x)\,dx + (2xyz + e^z)\,dy + (xy^2 + ye^z - 3z^2)\,dz$$

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4. **Method of Grouping Practice:** Find the scalar field $f(x, y)$ satisfying:

$$df = \frac{x\,dy - y\,dx}{x^2 + y^2} + \frac{x\,dx + y\,dy}{\sqrt{x^2 + y^2}}$$

5. **Systemic Implicit Differentiation:** Let u and v be implicitly defined as functions of x and y by:

$$uv = x, \quad u + v = y$$

Find $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$, and $\frac{\partial v}{\partial y}$.

6. **Second Order Implicit Surface Differentiation:** Let $z = f(x, y)$ be defined by $x^2 + y^2 + z^2 = r^2$. Show that:

$$\frac{\partial^2 z}{\partial x^2} = -\frac{r^2 - y^2}{z^3}$$

9 Comprehensive Collection of Solved Practice Problems

This section presents a curated set of 16 fully solved problems covering multivariable limits, continuity, differentiability, total differentials, chain rule transformations, implicit function systems, potential function reconstruction (in 2D and 3D), and integrating factor techniques.

Example 9.1 (Path-Dependence Limit Non-Existence Test)**.** Determine whether the following limit exists:

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2} \tag{99}$$

Solution. We test the behavior of $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ along different paths approaching the origin $(0, 0)$.

1. **Approach along linear paths $y = mx$:**

$$\lim_{x\to 0} f(x, mx) = \lim_{x\to 0} \frac{x^2(mx)}{x^4 + (mx)^2} = \lim_{x\to 0} \frac{mx^3}{x^4 + m^2x^2} = \lim_{x\to 0} \frac{mx}{x^2 + m^2} = 0 \tag{100}$$

Along every straight line $y = mx$, the limit appears to be 0.

2. **Approach along parabolic paths $y = kx^2$:**

$$\lim_{x\to 0} f(x, kx^2) = \lim_{x\to 0} \frac{x^2(kx^2)}{x^4 + (kx^2)^2} = \lim_{x\to 0} \frac{kx^4}{x^4 + k^2x^4} = \frac{k}{1 + k^2} \tag{101}$$

The limiting value depends explicitly on the choice of k:

- For $k = 1$ ($y = x^2$), the limit is $\frac{1}{1+1^2} = \frac{1}{2}$.
- For $k = 2$ ($y = 2x^2$), the limit is $\frac{2}{1+2^2} = \frac{2}{5}$.

By Theorem 1.1, since $f(x, y)$ yields different values along different continuous paths, the limit does not exist. □

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Example 9.2 (Limit Evaluation via Polar Coordinates)**.** Evaluate the limit or show that it does not exist:

$$\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} \tag{102}$$

Solution. Transform the expression to polar coordinates using $x = r\cos\theta$ and $y = r\sin\theta$, where $(x, y) \to (0, 0) \iff r \to 0^+$:

$$\frac{x^3 - y^3}{x^2 + y^2} = \frac{r^3\cos^3\theta - r^3\sin^3\theta}{r^2(\cos^2\theta + \sin^2\theta)} = r\left(\cos^3\theta - \sin^3\theta\right) \tag{103}$$

Establish an absolute upper bound using the triangle inequality:

$$\left|r\left(\cos^3\theta - \sin^3\theta\right)\right| \le r\left(|\cos^3\theta| + |\sin^3\theta|\right) \le 2r \tag{104}$$

Since $0 \le \left|\frac{x^3-y^3}{x^2+y^2}\right| \le 2r$ and $\lim_{r\to0^+} 2r = 0$, by the Squeeze Theorem:

$$\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} = 0 \tag{105}$$

□

Example 9.3 (Continuity Verification of a Piecewise Function)**.** Define $f : \mathbb{R}^2 \to \mathbb{R}$ by:

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases} \tag{106}$$

Determine whether f is continuous at $(0, 0)$.

Solution. To verify continuity at $(0, 0)$, we check if $\lim_{(x,y)\to(0,0)} f(x, y) = f(0, 0) = 0$.

Switching to polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$):

$$f(r\cos\theta, r\sin\theta) = \frac{(r\cos\theta)(r\sin\theta)}{\sqrt{r^2\cos^2\theta + r^2\sin^2\theta}} = \frac{r^2\cos\theta\sin\theta}{r} = r\cos\theta\sin\theta \tag{107}$$

Bounding the magnitude:

$$|f(r\cos\theta, r\sin\theta)| = r|\cos\theta\sin\theta| \le \frac{1}{2}r \xrightarrow{r\to0^+} 0 \tag{108}$$

Because $\lim_{(x,y)\to(0,0)} f(x, y) = 0 = f(0, 0)$, f is continuous at $(0, 0)$. □

Example 9.4 (Existence of Partial Derivatives vs. Total Differentiability)**.** Consider the function:

$$f(x, y) = \begin{cases} \frac{x^3}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases} \tag{109}$$

Show that the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ exist, but f is not totally differentiable at $(0, 0)$.

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Solution. **1. Compute partial derivatives at $(0,0)$ via limit definitions:**

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{h^3}{h^2+0} - 0}{h} = \lim_{h\to 0} \frac{h}{h} = 1 \tag{110}$$

$$f_y(0,0) = \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k\to 0} \frac{0-0}{k} = 0 \tag{111}$$

Both partial derivatives exist at $(0,0)$, with $\nabla f(0,0) = (1,0)$.

2. Apply the Total Differentiability Limit Condition: For f to be differentiable at $(0,0)$, the error term $E(h,k)$ must satisfy:

$$L = \lim_{(h,k)\to(0,0)} \frac{f(h,k) - f(0,0) - f_x(0,0)h - f_y(0,0)k}{\sqrt{h^2+k^2}} = 0 \tag{112}$$

Substitute the values:

$$L = \lim_{(h,k)\to(0,0)} \frac{\frac{h^3}{h^2+k^2} - h}{\sqrt{h^2+k^2}} = \lim_{(h,k)\to(0,0)} \frac{h^3 - h(h^2+k^2)}{(h^2+k^2)^{3/2}} = \lim_{(h,k)\to(0,0)} \frac{-hk^2}{(h^2+k^2)^{3/2}} \tag{113}$$

Test the approach along the line $k = mh$:

$$\lim_{h\to 0} \frac{-h(mh)^2}{(h^2+m^2h^2)^{3/2}} = \lim_{h\to 0} \frac{-m^2h^3}{|h|^3(1+m^2)^{3/2}} = \frac{-m^2\,\mathrm{sgn}(h)}{(1+m^2)^{3/2}} \tag{114}$$

Since this limit depends on m (and the sign of h), $L \neq 0$. Therefore, f is not differentiable at $(0,0)$. □

Example 9.5 (Verification of Clairaut's Theorem)**.** Verify Clairaut's Theorem ($f_{xy} = f_{yx}$) for $f(x,y) = \ln(x^2+y^2)$ on $\mathbb{R}^2 \setminus \{(0,0)\}$.

Solution. Compute first-order partial derivatives:

$$f_x = \frac{\partial}{\partial x}\ln(x^2+y^2) = \frac{2x}{x^2+y^2}, \quad f_y = \frac{\partial}{\partial y}\ln(x^2+y^2) = \frac{2y}{x^2+y^2} \tag{115}$$

Compute second-order mixed partial derivatives using the quotient rule:

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{2x}{x^2+y^2}\right) = \frac{0\cdot(x^2+y^2) - 2x(2y)}{(x^2+y^2)^2} = -\frac{4xy}{(x^2+y^2)^2} \tag{116}$$

$$f_{yx} = \frac{\partial}{\partial x}\left(\frac{2y}{x^2+y^2}\right) = \frac{0\cdot(x^2+y^2) - 2y(2x)}{(x^2+y^2)^2} = -\frac{4xy}{(x^2+y^2)^2} \tag{117}$$

Since $f_{xy} = f_{yx} = -\frac{4xy}{(x^2+y^2)^2}$ and both functions are continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$, Clairaut's Theorem holds. □

Example 9.6 (Linear Approximation and Total Differential)**.** Use the total differential of $f(x,y) = \sqrt{x^2+y^2}$ at $(3,4)$ to approximate $f(3.02, 3.99)$.

Solution. Set basepoint $(x_0, y_0) = (3,4)$ and increments $dx = 0.02$, $dy = -0.01$. At $(3,4)$, $f(3,4) = \sqrt{3^2+4^2} = 5$.

Compute partial derivatives at $(3,4)$:

$$f_x = \frac{x}{\sqrt{x^2+y^2}} \implies f_x(3,4) = \frac{3}{5} = 0.6 \tag{118}$$

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$$f_y = \frac{y}{\sqrt{x^2+y^2}} \implies f_y(3,4) = \frac{4}{5} = 0.8 \tag{119}$$

The total differential df is:

$$df = f_x dx + f_y dy = (0.6)(0.02) + (0.8)(-0.01) = 0.012 - 0.008 = 0.004 \tag{120}$$

The linear approximation gives:

$$f(3.02, 3.99) \approx f(3,4) + df = 5 + 0.004 = 5.004 \tag{121}$$

□

Example 9.7 (Single-Parameter Chain Rule Evaluation)**.** Let $z = e^x \sin y$, where $x = t^2$ and $y = \ln t$. Compute $\frac{dz}{dt}$ at $t = 1$.

Solution. At $t = 1$, we have $x = 1^2 = 1$ and $y = \ln(1) = 0$.

Compute partial derivatives of z with respect to x and y:

$$\frac{\partial z}{\partial x} = e^x \sin y \implies \left.\frac{\partial z}{\partial x}\right|_{(1,0)} = e^1 \sin(0) = 0 \tag{122}$$

$$\frac{\partial z}{\partial y} = e^x \cos y \implies \left.\frac{\partial z}{\partial y}\right|_{(1,0)} = e^1 \cos(0) = e \tag{123}$$

Compute ordinary derivatives of x and y with respect to t:

$$\frac{dx}{dt} = 2t \implies \left.\frac{dx}{dt}\right|_{t=1} = 2 \tag{124}$$

$$\frac{dy}{dt} = \frac{1}{t} \implies \left.\frac{dy}{dt}\right|_{t=1} = 1 \tag{125}$$

Apply the single-parameter Chain Rule:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} = (0)(2) + (e)(1) = e \tag{126}$$

□

Example 9.8 (Laplacian Transformation to Polar Coordinates)**.** Let $u(x, y)$ be C^2. Under the polar transformation $x = r\cos\theta, y = r\sin\theta$, prove that the Laplacian $\Delta u = u_{xx} + u_{yy}$ transforms to:

$$u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} \tag{127}$$

Solution. Using $r = \sqrt{x^2+y^2}$ and $\theta = \arctan(y/x)$, compute partials:

$$r_x = \frac{x}{r} = \cos\theta, \quad r_y = \frac{y}{r} = \sin\theta, \quad \theta_x = -\frac{\sin\theta}{r}, \quad \theta_y = \frac{\cos\theta}{r} \tag{128}$$

Apply the chain rule operator for $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$:

$$\frac{\partial}{\partial x} = \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta} \tag{129}$$

$$\frac{\partial}{\partial y} = \sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial\theta} \tag{130}$$

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Applying $\frac{\partial}{\partial x}$ to $u_x = \cos\theta u_r - \frac{\sin\theta}{r}u_\theta$:

$$u_{xx} = \left(\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right)\left(\cos\theta u_r - \frac{\sin\theta}{r}u_\theta\right) \tag{131}$$

Expanding gives:

$$u_{xx} = \cos^2\theta u_{rr} - \frac{2\sin\theta\cos\theta}{r}u_{r\theta} + \frac{\sin^2\theta}{r^2}u_{\theta\theta} + \frac{\sin^2\theta}{r}u_r + \frac{2\sin\theta\cos\theta}{r^2}u_\theta \tag{132}$$

Similarly, expanding u_{yy}:

$$u_{yy} = \sin^2\theta u_{rr} + \frac{2\sin\theta\cos\theta}{r}u_{r\theta} + \frac{\cos^2\theta}{r^2}u_{\theta\theta} + \frac{\cos^2\theta}{r}u_r - \frac{2\sin\theta\cos\theta}{r^2}u_\theta \tag{133}$$

Summing u_{xx} and u_{yy} using $\cos^2\theta + \sin^2\theta = 1$:

$$u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} \tag{134}$$

□

Example 9.9 (Implicit Partial Derivative for a Single Relation)**.** Given $e^{xyz} + x^2 + y^2 + z^2 = 3$, calculate $\frac{\partial z}{\partial x}$ at $(1, 1, 0)$.

Solution. Define $F(x, y, z) = e^{xyz} + x^2 + y^2 + z^2 - 3 = 0$.

Compute partial derivatives of F:

$$F_x = yze^{xyz} + 2x \tag{135}$$
$$F_z = xye^{xyz} + 2z \tag{136}$$

Evaluate F_x and F_z at $P(1, 1, 0)$:

$$F_x(1, 1, 0) = (1)(0)e^0 + 2(1) = 2 \tag{137}$$
$$F_z(1, 1, 0) = (1)(1)e^0 + 2(0) = 1 \neq 0 \tag{138}$$

By Theorem 4.1:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2}{1} = -2 \tag{139}$$

□

Example 9.10 (System of Implicit Functions via Jacobian Determinants)**.** Let u and v be implicitly defined as functions of x and y by:

$$F(x, y, u, v) = u + v - x - y = 0 \tag{140}$$
$$G(x, y, u, v) = xu + yv - 1 = 0 \tag{141}$$

Find $\frac{\partial u}{\partial x}$ using Jacobian matrices.

Solution. Compute the Jacobian matrix of (F, G) with respect to (u, v):

$$J_{(u,v)} = \begin{bmatrix} F_u & F_v \\ G_u & G_v \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ x & y \end{bmatrix} \implies \det(J_{(u,v)}) = y - x \tag{142}$$

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Provided $y \neq x$, u and v are differentiable. Differentiating the system with respect to x:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} - 1 = 0 \implies \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} = 1 \tag{143}$$

$$u + x\frac{\partial u}{\partial x} + y\frac{\partial v}{\partial x} = 0 \implies x\frac{\partial u}{\partial x} + y\frac{\partial v}{\partial x} = -u \tag{144}$$

In matrix form:

$$\begin{bmatrix} 1 & 1 \\ x & y \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} 1 \\ -u \end{bmatrix} \tag{145}$$

Applying Cramer's Rule for $\frac{\partial u}{\partial x}$:

$$\frac{\partial u}{\partial x} = \frac{\begin{vmatrix} 1 & 1 \\ -u & y \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ x & y \end{vmatrix}} = \frac{y - (-u)}{y - x} = \frac{u + y}{y - x} \tag{146}$$

□

Example 9.11 (2D Potential Function Reconstruction with Mixed Exponential Terms)**.** Determine whether the form $\omega = (e^y + ye^x + 2x)\,dx + (xe^y + e^x - 3y^2)\,dy$ is exact. If exact, reconstruct its potential function $f(x, y)$.

Solution. **Step 1: Check Exactness Criterion.** $M(x, y) = e^y + ye^x + 2x$ and $N(x, y) = xe^y + e^x - 3y^2$.

$$\frac{\partial M}{\partial y} = e^y + e^x \tag{147}$$

$$\frac{\partial N}{\partial x} = e^y + e^x \tag{148}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, ω is exact.

Step 2: Partial Integration w.r.t. x.

$$f(x, y) = \int M(x, y)\,dx + g(y) = \int (e^y + ye^x + 2x)\,dx + g(y) = xe^y + ye^x + x^2 + g(y) \tag{149}$$

Step 3: Differentiate w.r.t. y and match with $N(x, y)$.

$$\frac{\partial f}{\partial y} = xe^y + e^x + g'(y) = xe^y + e^x - 3y^2 \implies g'(y) = -3y^2 \tag{150}$$

Integrating $g'(y)$ gives $g(y) = -y^3 + C$.

Final Result:

$$f(x, y) = xe^y + ye^x + x^2 - y^3 + C \tag{151}$$

□

Example 9.12 (3D Potential Function Reconstruction with Rational Terms)**.** Reconstruct the scalar potential $f(x, y, z)$ for the differential:

$$df = \left(\frac{1}{y} + 2xz\right) dx + \left(ze^y - \frac{x}{y^2}\right) dy + \left(e^y + x^2\right) dz \tag{152}$$

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Solution. **Step 1: Check 3D Exactness.** $M = \frac{1}{y} + 2xz$, $N = ze^y - \frac{x}{y^2}$, $P = e^y + x^2$.

$$M_y = -\frac{1}{y^2} = N_x, \tag{153}$$

$$N_z = e^y = P_y, \tag{154}$$

$$P_x = 2x = M_z \tag{155}$$

The field is exact.

Step 2: Sequential Integration. 1. Integrate M with respect to x:

$$f(x,y,z) = \int \left(\frac{1}{y} + 2xz\right) dx = \frac{x}{y} + x^2 z + g(y,z) \tag{156}$$

2. Differentiate f with respect to y and set equal to N:

$$\frac{\partial f}{\partial y} = -\frac{x}{y^2} + \frac{\partial g}{\partial y} = ze^y - \frac{x}{y^2} \implies \frac{\partial g}{\partial y} = ze^y \implies g(y,z) = ze^y + h(z) \tag{157}$$

3. Differentiate $f = \frac{x}{y} + x^2 z + ze^y + h(z)$ with respect to z and set equal to P:

$$\frac{\partial f}{\partial z} = x^2 + e^y + h'(z) = e^y + x^2 \implies h'(z) = 0 \implies h(z) = C \tag{158}$$

Final Result:

$$f(x,y,z) = \frac{x}{y} + x^2 z + ze^y + C \tag{159}$$

□

Example 9.13 (Definite Path Integration Method for Potential Reconstruction)**.** Reconstruct $f(x,y)$ such that $df = (2xy + 3)\,dx + (x^2 - 4y)\,dy$ with $f(0,0) = 1$ using path-independent definite line integration.

Solution. Since $M_y = 2x = N_x$, the form is exact. Using the path integration formula from Section 5.2 from basepoint $(x_0, y_0) = (0,0)$:

$$f(x,y) = \int_0^x M(t,0)\,dt + \int_0^y N(x,t)\,dt + C \tag{160}$$

Compute the two component integrals:

$$\int_0^x M(t,0)\,dt = \int_0^x (2t(0) + 3)\,dt = \int_0^x 3\,dt = 3x \tag{161}$$

$$\int_0^y N(x,t)\,dt = \int_0^y (x^2 - 4t)\,dt = \left[x^2 t - 2t^2\right]_0^y = x^2 y - 2y^2 \tag{162}$$

Combining both parts:

$$f(x,y) = 3x + x^2 y - 2y^2 + C \tag{163}$$

Apply initial condition $f(0,0) = 1 \implies C = 1$:

$$f(x,y) = x^2 y + 3x - 2y^2 + 1 \tag{164}$$

□

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Example 9.14 (Single-Variable Integrating Factor $\mu(y)$)**.** Find an integrating factor depending only on y for:

$$y\,dx + (2x - ye^y)\,dy = 0 \tag{165}$$

and determine the corresponding exact potential function.

Solution. **1. Test Exactness:** $M(x,y) = y$, $N(x,y) = 2x - ye^y$.

$$M_y = 1, \quad N_x = 2 \implies M_y \neq N_x \quad \text{(Inexact)} \tag{166}$$

2. Find Integrating Factor $\mu(y)$: Apply Lemma 7.3:

$$g(y) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{2-1}{y} = \frac{1}{y} \tag{167}$$

Since $g(y)$ depends purely on y:

$$\mu(y) = \exp\left(\int \frac{1}{y}\,dy\right) = e^{\ln y} = y \tag{168}$$

3. Multiply Equation by $\mu(y) = y$ and Solve:

$$y^2\,dx + (2xy - y^2e^y)\,dy = 0 \tag{169}$$

Integrate $M_{\text{new}} = y^2$ with respect to x:

$$f(x,y) = \int y^2\,dx = xy^2 + g(y) \tag{170}$$

Differentiate with respect to y and match $N_{\text{new}} = 2xy - y^2e^y$:

$$\frac{\partial f}{\partial y} = 2xy + g'(y) = 2xy - y^2e^y \implies g'(y) = -y^2e^y \tag{171}$$

Integrating $g'(y)$ by parts twice:

$$g(y) = \int -y^2e^y\,dy = -y^2e^y + 2ye^y - 2e^y + C \tag{172}$$

Thus, the potential function is:

$$f(x,y) = xy^2 - e^y(y^2 - 2y + 2) + C \tag{173}$$

□

Example 9.15 (Product Variable Integrating Factor $\mu(xy)$)**.** Solve the inexact differential form using an integrating factor of the form $\mu = \mu(z)$ where $z = xy$:

$$(y + x^2y^3)\,dx + (x - x^3y^2)\,dy = 0 \tag{174}$$

Solution. **1. Test Exactness:** $M = y + x^2y^3$, $N = x - x^3y^2$.

$$M_y = 1 + 3x^2y^2, \quad N_x = 1 - 3x^2y^2 \implies M_y - N_x = 6x^2y^2 \neq 0 \tag{175}$$

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2. Find Integrating Factor for $z = xy$: Apply Corollary 7.5:

$$h(z) = \frac{M_y - N_x}{yN - xM} = \frac{6x^2y^2}{y(x - x^3y^2) - x(y + x^2y^3)} = \frac{6x^2y^2}{-2x^3y^3} = -\frac{3}{xy} = -\frac{3}{z} \tag{176}$$

Since $h(z)$ depends purely on z:

$$\mu(z) = \exp\left(\int -\frac{3}{z}\, dz\right) = z^{-3} = \frac{1}{x^3y^3} \tag{177}$$

3. Multiply by $\mu = \frac{1}{x^3y^3}$:

$$\left(\frac{1}{x^3y^2} + \frac{1}{x}\right) dx + \left(\frac{1}{x^2y^3} - \frac{1}{y}\right) dy = 0 \tag{178}$$

Integrate M_{new} with respect to x:

$$f(x, y) = \int \left(x^{-3}y^{-2} + \frac{1}{x}\right) dx = -\frac{1}{2x^2y^2} + \ln|x| + g(y) \tag{179}$$

Differentiate w.r.t. y:

$$\frac{\partial f}{\partial y} = \frac{1}{x^2y^3} + g'(y) = \frac{1}{x^2y^3} - \frac{1}{y} \implies g'(y) = -\frac{1}{y} \implies g(y) = -\ln|y| \tag{180}$$

Final Result:

$$f(x, y) = \ln\left|\frac{x}{y}\right| - \frac{1}{2x^2y^2} + C \tag{181}$$

□

Example 9.16 (Homogeneous Differential Form Integrating Factor)**.** Find an integrating factor for the homogeneous differential form:

$$(x^2y - 2xy^2)\, dx - (x^3 - 3x^2y)\, dy = 0 \tag{182}$$

and transform it into an exact differential.

Solution. **1. Verify Homogeneity:** $M(x, y) = x^2y - 2xy^2$ and $N(x, y) = -x^3 + 3x^2y$ are homogeneous of degree $n = 3$.

2. Apply Euler's Theorem Criterion for Homogeneous Forms: Compute the denominator $D(x, y) = xM + yN$:

$$\begin{aligned} xM + yN &= x(x^2y - 2xy^2) + y(-x^3 + 3x^2y) \\ &= x^3y - 2x^2y^2 - x^3y + 3x^2y^2 = x^2y^2 \end{aligned} \tag{183}$$

Since $xM + yN = x^2y^2 \neq 0$, Theorem 7.6 guarantees that the integrating factor is:

$$\mu(x, y) = \frac{1}{xM + yN} = \frac{1}{x^2y^2} \tag{184}$$

3. Multiply and Verify Exactness:

$$\omega_\mu = \left(\frac{1}{y} - \frac{2}{x}\right) dx + \left(\frac{3}{y} - \frac{x}{y^2}\right) dy \tag{185}$$

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Verify exactness of modified components:

$$\frac{\partial}{\partial y}\left(\frac{1}{y}-\frac{2}{x}\right)=-\frac{1}{y^2},\quad \frac{\partial}{\partial x}\left(\frac{3}{y}-\frac{x}{y^2}\right)=-\frac{1}{y^2}\quad\checkmark \tag{186}$$

Reconstructing potential $f(x,y)$:

$$f(x,y)=\int\left(\frac{1}{y}-\frac{2}{x}\right)dx=\frac{x}{y}-2\ln|x|+g(y) \tag{187}$$

$$\frac{\partial f}{\partial y}=-\frac{x}{y^2}+g'(y)=\frac{3}{y}-\frac{x}{y^2}\implies g'(y)=\frac{3}{y}\implies g(y)=3\ln|y| \tag{188}$$

Final Result:

$$f(x,y)=\frac{x}{y}+\ln\left|\frac{y^3}{x^2}\right|+C \tag{189}$$

□

10 Limits and Continuity

Example 10.1 (Path-Dependence Limit Non-Existence Test)**.** Determine whether the following limit exists:

$$\lim_{(x,y)\to(0,0)}\frac{x^2y}{x^4+y^2} \tag{190}$$

Solution. We test the behavior of $f(x,y)=\frac{x^2y}{x^4+y^2}$ along different paths approaching the origin $(0,0)$.

1. **Approach along linear paths $y=mx$:**

$$\lim_{x\to0}f(x,mx)=\lim_{x\to0}\frac{x^2(mx)}{x^4+(mx)^2}=\lim_{x\to0}\frac{mx^3}{x^4+m^2x^2}=\lim_{x\to0}\frac{mx}{x^2+m^2}=0 \tag{191}$$

Along every straight line $y=mx$, the limit appears to be 0.

2. **Approach along parabolic paths $y=kx^2$:**

$$\lim_{x\to0}f(x,kx^2)=\lim_{x\to0}\frac{x^2(kx^2)}{x^4+(kx^2)^2}=\lim_{x\to0}\frac{kx^4}{x^4+k^2x^4}=\frac{k}{1+k^2} \tag{192}$$

The limiting value depends explicitly on the choice of k:

- For $k=1$ ($y=x^2$), the limit is $\frac{1}{1+1^2}=\frac{1}{2}$.
- For $k=2$ ($y=2x^2$), the limit is $\frac{2}{1+2^2}=\frac{2}{5}$.

Since $f(x,y)$ yields different values along different continuous paths, the limit does not exist. □

Example 10.2 (Limit Evaluation via Polar Coordinates)**.** Evaluate the limit or show that it does not exist:

$$\lim_{(x,y)\to(0,0)}\frac{x^3-y^3}{x^2+y^2} \tag{193}$$

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Solution. Transform the expression to polar coordinates using $x = r\cos\theta$ and $y = r\sin\theta$, where $(x, y) \to (0, 0) \iff r \to 0^+$:

$$\frac{x^3 - y^3}{x^2 + y^2} = \frac{r^3\cos^3\theta - r^3\sin^3\theta}{r^2(\cos^2\theta + \sin^2\theta)} = r\left(\cos^3\theta - \sin^3\theta\right) \tag{194}$$

Establish an absolute upper bound using the triangle inequality:

$$\left|r\left(\cos^3\theta - \sin^3\theta\right)\right| \le r\left(|\cos^3\theta| + |\sin^3\theta|\right) \le 2r \tag{195}$$

Since $0 \le \left|\frac{x^3-y^3}{x^2+y^2}\right| \le 2r$ and $\lim_{r\to 0^+} 2r = 0$, by the Squeeze Theorem:

$$\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} = 0 \tag{196}$$

□

Example 10.3 (Continuity Verification of a Piecewise Function)**.** Define $f : \mathbb{R}^2 \to \mathbb{R}$ by:

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases} \tag{197}$$

Determine whether f is continuous at $(0, 0)$.

Solution. To verify continuity at $(0, 0)$, we check if $\lim_{(x,y)\to(0,0)} f(x, y) = f(0, 0) = 0$.

Switching to polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$):

$$f(r\cos\theta, r\sin\theta) = \frac{(r\cos\theta)(r\sin\theta)}{\sqrt{r^2\cos^2\theta + r^2\sin^2\theta}} = \frac{r^2\cos\theta\sin\theta}{r} = r\cos\theta\sin\theta \tag{198}$$

Bounding the magnitude:

$$|f(r\cos\theta, r\sin\theta)| = r|\cos\theta\sin\theta| \le \frac{1}{2}r \xrightarrow{r\to 0^+} 0 \tag{199}$$

Because $\lim_{(x,y)\to(0,0)} f(x, y) = 0 = f(0, 0)$, f is continuous at $(0, 0)$. □

11 Partial Derivatives and Differentiability

Example 11.1 (Existence of Partial Derivatives vs. Total Differentiability)**.** Consider the function:

$$f(x, y) = \begin{cases} \frac{x^3}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases} \tag{200}$$

Show that the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ exist, but f is not totally differentiable at $(0, 0)$.

Solution. **1. Compute partial derivatives at** $(0, 0)$ **via limit definitions:**

$$f_x(0, 0) = \lim_{h\to 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h\to 0} \frac{\frac{h^3}{h^2+0} - 0}{h} = \lim_{h\to 0} \frac{h}{h} = 1 \tag{201}$$

$$f_y(0, 0) = \lim_{k\to 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k\to 0} \frac{0 - 0}{k} = 0 \tag{202}$$

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Both partial derivatives exist at $(0,0)$, with $\nabla f(0,0) = (1,0)$.

2. Apply the Total Differentiability Limit Condition: For f to be differentiable at $(0,0)$, the error term $E(h,k)$ must satisfy:

$$L = \lim_{(h,k)\to(0,0)} \frac{f(h,k) - f(0,0) - f_x(0,0)h - f_y(0,0)k}{\sqrt{h^2+k^2}} = 0 \tag{203}$$

Substitute the values:

$$L = \lim_{(h,k)\to(0,0)} \frac{\frac{h^3}{h^2+k^2} - h}{\sqrt{h^2+k^2}} = \lim_{(h,k)\to(0,0)} \frac{h^3 - h(h^2+k^2)}{(h^2+k^2)^{3/2}} = \lim_{(h,k)\to(0,0)} \frac{-hk^2}{(h^2+k^2)^{3/2}} \tag{204}$$

Test the approach along the line $k = mh$:

$$\lim_{h\to 0} \frac{-h(mh)^2}{(h^2+m^2h^2)^{3/2}} = \lim_{h\to 0} \frac{-m^2h^3}{|h|^3(1+m^2)^{3/2}} = \frac{-m^2\,\mathrm{sgn}(h)}{(1+m^2)^{3/2}} \tag{205}$$

Since this limit depends on m (and the sign of h), $L \neq 0$. Therefore, f is not differentiable at $(0,0)$. □

Example 11.2 (Verification of Clairaut's Theorem)**.** Verify Clairaut's Theorem ($f_{xy} = f_{yx}$) for $f(x,y) = \ln(x^2+y^2)$ on $\mathbb{R}^2 \setminus \{(0,0)\}$.

Solution. Compute first-order partial derivatives:

$$f_x = \frac{\partial}{\partial x}\ln(x^2+y^2) = \frac{2x}{x^2+y^2}, \quad f_y = \frac{\partial}{\partial y}\ln(x^2+y^2) = \frac{2y}{x^2+y^2} \tag{206}$$

Compute second-order mixed partial derivatives using the quotient rule:

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{2x}{x^2+y^2}\right) = \frac{0\cdot(x^2+y^2) - 2x(2y)}{(x^2+y^2)^2} = -\frac{4xy}{(x^2+y^2)^2} \tag{207}$$

$$f_{yx} = \frac{\partial}{\partial x}\left(\frac{2y}{x^2+y^2}\right) = \frac{0\cdot(x^2+y^2) - 2y(2x)}{(x^2+y^2)^2} = -\frac{4xy}{(x^2+y^2)^2} \tag{208}$$

Since $f_{xy} = f_{yx} = -\frac{4xy}{(x^2+y^2)^2}$ and both functions are continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$, Clairaut's Theorem holds. □

Example 11.3 (Linear Approximation and Total Differential)**.** Use the total differential of $f(x,y) = \sqrt{x^2+y^2}$ at $(3,4)$ to approximate $f(3.02, 3.99)$.

Solution. Set basepoint $(x_0, y_0) = (3,4)$ and increments $dx = 0.02$, $dy = -0.01$. At $(3,4)$, $f(3,4) = \sqrt{3^2+4^2} = 5$.

Compute partial derivatives at $(3,4)$:

$$f_x = \frac{x}{\sqrt{x^2+y^2}} \implies f_x(3,4) = \frac{3}{5} = 0.6 \tag{209}$$

$$f_y = \frac{y}{\sqrt{x^2+y^2}} \implies f_y(3,4) = \frac{4}{5} = 0.8 \tag{210}$$

The total differential df is:

$$df = f_x dx + f_y dy = (0.6)(0.02) + (0.8)(-0.01) = 0.012 - 0.008 = 0.004 \tag{211}$$

The linear approximation gives:

$$f(3.02, 3.99) \approx f(3,4) + df = 5 + 0.004 = 5.004 \tag{212}$$

□

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12 Multivariable Chain Rule

Example 12.1 (Single-Parameter Chain Rule Evaluation)**.** Let $z = e^x \sin y$, where $x = t^2$ and $y = \ln t$. Compute $\frac{dz}{dt}$ at $t = 1$.

Solution. At $t = 1$, we have $x = 1^2 = 1$ and $y = \ln(1) = 0$.

Compute partial derivatives of z with respect to x and y:

$$\frac{\partial z}{\partial x} = e^x \sin y \implies \left.\frac{\partial z}{\partial x}\right|_{(1,0)} = e^1 \sin(0) = 0 \tag{213}$$

$$\frac{\partial z}{\partial y} = e^x \cos y \implies \left.\frac{\partial z}{\partial y}\right|_{(1,0)} = e^1 \cos(0) = e \tag{214}$$

Compute ordinary derivatives of x and y with respect to t:

$$\frac{dx}{dt} = 2t \implies \left.\frac{dx}{dt}\right|_{t=1} = 2 \tag{215}$$

$$\frac{dy}{dt} = \frac{1}{t} \implies \left.\frac{dy}{dt}\right|_{t=1} = 1 \tag{216}$$

Apply the single-parameter Chain Rule:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} = (0)(2) + (e)(1) = e \tag{217}$$

□

Example 12.2 (Laplacian Transformation to Polar Coordinates)**.** Let $u(x, y)$ be C^2. Under the polar transformation $x = r\cos\theta, y = r\sin\theta$, prove that the Laplacian $\Delta u = u_{xx} + u_{yy}$ transforms to:

$$u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} \tag{218}$$

Solution. Using $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$, compute partials:

$$r_x = \frac{x}{r} = \cos\theta, \quad r_y = \frac{y}{r} = \sin\theta, \quad \theta_x = -\frac{\sin\theta}{r}, \quad \theta_y = \frac{\cos\theta}{r} \tag{219}$$

Apply the chain rule operator for $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$:

$$\frac{\partial}{\partial x} = \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta} \tag{220}$$

$$\frac{\partial}{\partial y} = \sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial\theta} \tag{221}$$

Applying $\frac{\partial}{\partial x}$ to $u_x = \cos\theta u_r - \frac{\sin\theta}{r}u_\theta$:

$$u_{xx} = \left(\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right)\left(\cos\theta u_r - \frac{\sin\theta}{r}u_\theta\right) \tag{222}$$

Expanding gives:

$$u_{xx} = \cos^2\theta u_{rr} - \frac{2\sin\theta\cos\theta}{r}u_{r\theta} + \frac{\sin^2\theta}{r^2}u_{\theta\theta} + \frac{\sin^2\theta}{r}u_r + \frac{2\sin\theta\cos\theta}{r^2}u_\theta \tag{223}$$

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Similarly, expanding u_{yy}:

$$u_{yy} = \sin^2\theta u_{rr} + \frac{2\sin\theta\cos\theta}{r}u_{r\theta} + \frac{\cos^2\theta}{r^2}u_{\theta\theta} + \frac{\cos^2\theta}{r}u_r - \frac{2\sin\theta\cos\theta}{r^2}u_\theta \tag{224}$$

Summing u_{xx} and u_{yy} using $\cos^2\theta + \sin^2\theta = 1$:

$$u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} \tag{225}$$

□

13 Implicit Functions

Example 13.1 (Implicit Partial Derivative for a Single Relation)**.** Given $e^{xyz} + x^2 + y^2 + z^2 = 3$, calculate $\frac{\partial z}{\partial x}$ at $(1, 1, 0)$.

Solution. Define $F(x, y, z) = e^{xyz} + x^2 + y^2 + z^2 - 3 = 0$.

Compute partial derivatives of F:

$$F_x = yze^{xyz} + 2x \tag{226}$$
$$F_z = xye^{xyz} + 2z \tag{227}$$

Evaluate F_x and F_z at $P(1, 1, 0)$:

$$F_x(1, 1, 0) = (1)(0)e^0 + 2(1) = 2 \tag{228}$$
$$F_z(1, 1, 0) = (1)(1)e^0 + 2(0) = 1 \neq 0 \tag{229}$$

By the Implicit Function Theorem:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2}{1} = -2 \tag{230}$$

□

Example 13.2 (System of Implicit Functions via Jacobian Determinants)**.** Let u and v be implicitly defined as functions of x and y by:

$$F(x, y, u, v) = u + v - x - y = 0 \tag{231}$$
$$G(x, y, u, v) = xu + yv - 1 = 0 \tag{232}$$

Find $\frac{\partial u}{\partial x}$ using Jacobian matrices.

Solution. Compute the Jacobian matrix of (F, G) with respect to (u, v):

$$J_{(u,v)} = \begin{bmatrix} F_u & F_v \\ G_u & G_v \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ x & y \end{bmatrix} \implies \det(J_{(u,v)}) = y - x \tag{233}$$

Provided $y \neq x$, u and v are differentiable. Differentiating the system with respect to x:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} - 1 = 0 \implies \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} = 1 \tag{234}$$
$$u + x\frac{\partial u}{\partial x} + y\frac{\partial v}{\partial x} = 0 \implies x\frac{\partial u}{\partial x} + y\frac{\partial v}{\partial x} = -u \tag{235}$$

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In matrix form:

$$\begin{bmatrix} 1 & 1 \\ x & y \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} 1 \\ -u \end{bmatrix} \tag{236}$$

Applying Cramer's Rule for $\frac{\partial u}{\partial x}$:

$$\frac{\partial u}{\partial x} = \frac{\begin{vmatrix} 1 & 1 \\ -u & y \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ x & y \end{vmatrix}} = \frac{y-(-u)}{y-x} = \frac{u+y}{y-x} \tag{237}$$

□

14 Exact Differentials and Potential Reconstruction

Example 14.1 (2D Potential Function Reconstruction with Mixed Exponential Terms)**.** Determine whether the form $\omega = (e^y + ye^x + 2x)\,dx + (xe^y + e^x - 3y^2)\,dy$ is exact. If exact, reconstruct its potential function $f(x, y)$.

Solution. **Step 1: Check Exactness Criterion.** $M(x, y) = e^y + ye^x + 2x$ and $N(x, y) = xe^y + e^x - 3y^2$.

$$\frac{\partial M}{\partial y} = e^y + e^x \tag{238}$$

$$\frac{\partial N}{\partial x} = e^y + e^x \tag{239}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, ω is exact.

Step 2: Partial Integration w.r.t. x.

$$f(x, y) = \int M(x, y)\,dx + g(y) = \int (e^y + ye^x + 2x)\,dx + g(y) = xe^y + ye^x + x^2 + g(y) \tag{240}$$

Step 3: Differentiate w.r.t. y and match with $N(x, y)$.

$$\frac{\partial f}{\partial y} = xe^y + e^x + g'(y) = xe^y + e^x - 3y^2 \implies g'(y) = -3y^2 \tag{241}$$

Integrating $g'(y)$ gives $g(y) = -y^3 + C$.

Final Result:

$$f(x, y) = xe^y + ye^x + x^2 - y^3 + C \tag{242}$$

□

Example 14.2 (3D Potential Function Reconstruction with Rational Terms)**.** Reconstruct the scalar potential $f(x, y, z)$ for the differential:

$$df = \left(\frac{1}{y} + 2xz\right) dx + \left(ze^y - \frac{x}{y^2}\right) dy + \left(e^y + x^2\right) dz \tag{243}$$

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Solution. **Step 1: Check 3D Exactness.** $M = \frac{1}{y} + 2xz$, $N = ze^y - \frac{x}{y^2}$, $P = e^y + x^2$.

$$M_y = -\frac{1}{y^2} = N_x, \tag{244}$$

$$N_z = e^y = P_y, \tag{245}$$

$$P_x = 2x = M_z \tag{246}$$

The field is exact.

Step 2: Sequential Integration. 1. Integrate M with respect to x:

$$f(x, y, z) = \int \left(\frac{1}{y} + 2xz\right) dx = \frac{x}{y} + x^2 z + g(y, z) \tag{247}$$

2. Differentiate f with respect to y and set equal to N:

$$\frac{\partial f}{\partial y} = -\frac{x}{y^2} + \frac{\partial g}{\partial y} = ze^y - \frac{x}{y^2} \implies \frac{\partial g}{\partial y} = ze^y \implies g(y, z) = ze^y + h(z) \tag{248}$$

3. Differentiate $f = \frac{x}{y} + x^2 z + ze^y + h(z)$ with respect to z and set equal to P:

$$\frac{\partial f}{\partial z} = x^2 + e^y + h'(z) = e^y + x^2 \implies h'(z) = 0 \implies h(z) = C \tag{249}$$

Final Result:

$$f(x, y, z) = \frac{x}{y} + x^2 z + ze^y + C \tag{250}$$

□

Example 14.3 (Definite Path Integration Method for Potential Reconstruction)**.** Reconstruct $f(x, y)$ such that $df = (2xy + 3)\,dx + (x^2 - 4y)\,dy$ with $f(0, 0) = 1$ using path-independent definite line integration.

Solution. Since $M_y = 2x = N_x$, the form is exact. Using path integration from basepoint $(x_0, y_0) = (0, 0)$:

$$f(x, y) = \int_0^x M(t, 0)\,dt + \int_0^y N(x, t)\,dt + C \tag{251}$$

Compute the two component integrals:

$$\int_0^x M(t, 0)\,dt = \int_0^x (2t(0) + 3)\,dt = \int_0^x 3\,dt = 3x \tag{252}$$

$$\int_0^y N(x, t)\,dt = \int_0^y (x^2 - 4t)\,dt = \left[x^2 t - 2t^2\right]_0^y = x^2 y - 2y^2 \tag{253}$$

Combining both parts:

$$f(x, y) = 3x + x^2 y - 2y^2 + C \tag{254}$$

Apply initial condition $f(0, 0) = 1 \implies C = 1$:

$$f(x, y) = x^2 y + 3x - 2y^2 + 1 \tag{255}$$

□

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15 Integrating Factors

Example 15.1 (Single-Variable Integrating Factor $\mu(y)$)**.** Find an integrating factor depending only on y for:

$$y\,dx + (2x - ye^y)\,dy = 0 \tag{256}$$

and determine the corresponding exact potential function.

Solution. **1. Test Exactness:** $M(x,y) = y$, $N(x,y) = 2x - ye^y$.

$$M_y = 1, \quad N_x = 2 \implies M_y \neq N_x \quad \text{(Inexact)} \tag{257}$$

2. Find Integrating Factor $\mu(y)$:

$$g(y) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{2-1}{y} = \frac{1}{y} \tag{258}$$

Since $g(y)$ depends purely on y:

$$\mu(y) = \exp\left(\int \frac{1}{y}\,dy\right) = e^{\ln y} = y \tag{259}$$

3. Multiply Equation by $\mu(y) = y$ and Solve:

$$y^2\,dx + (2xy - y^2e^y)\,dy = 0 \tag{260}$$

Integrate $M_{\text{new}} = y^2$ with respect to x:

$$f(x,y) = \int y^2\,dx = xy^2 + g(y) \tag{261}$$

Differentiate with respect to y and match $N_{\text{new}} = 2xy - y^2e^y$:

$$\frac{\partial f}{\partial y} = 2xy + g'(y) = 2xy - y^2e^y \implies g'(y) = -y^2e^y \tag{262}$$

Integrating $g'(y)$ by parts twice:

$$g(y) = \int -y^2e^y\,dy = -y^2e^y + 2ye^y - 2e^y + C \tag{263}$$

Thus, the potential function is:

$$f(x,y) = xy^2 - e^y(y^2 - 2y + 2) + C \tag{264}$$

□

Example 15.2 (Product Variable Integrating Factor $\mu(xy)$)**.** Solve the inexact differential form using an integrating factor of the form $\mu = \mu(z)$ where $z = xy$:

$$(y + x^2y^3)\,dx + (x - x^3y^2)\,dy = 0 \tag{265}$$

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Solution. **1. Test Exactness:** $M = y + x^2y^3$, $N = x - x^3y^2$.

$$M_y = 1 + 3x^2y^2, \quad N_x = 1 - 3x^2y^2 \implies M_y - N_x = 6x^2y^2 \neq 0 \tag{266}$$

2. Find Integrating Factor for $z = xy$:

$$h(z) = \frac{M_y - N_x}{yN - xM} = \frac{6x^2y^2}{y(x - x^3y^2) - x(y + x^2y^3)} = \frac{6x^2y^2}{-2x^3y^3} = -\frac{3}{xy} = -\frac{3}{z} \tag{267}$$

Since $h(z)$ depends purely on z:

$$\mu(z) = \exp\left(\int -\frac{3}{z}\, dz\right) = z^{-3} = \frac{1}{x^3y^3} \tag{268}$$

3. Multiply by $\mu = \frac{1}{x^3y^3}$:

$$\left(\frac{1}{x^3y^2} + \frac{1}{x}\right) dx + \left(\frac{1}{x^2y^3} - \frac{1}{y}\right) dy = 0 \tag{269}$$

Integrate M_{new} with respect to x:

$$f(x, y) = \int \left(x^{-3}y^{-2} + \frac{1}{x}\right) dx = -\frac{1}{2x^2y^2} + \ln|x| + g(y) \tag{270}$$

Differentiate w.r.t. y:

$$\frac{\partial f}{\partial y} = \frac{1}{x^2y^3} + g'(y) = \frac{1}{x^2y^3} - \frac{1}{y} \implies g'(y) = -\frac{1}{y} \implies g(y) = -\ln|y| \tag{271}$$

Final Result:

$$f(x, y) = \ln\left|\frac{x}{y}\right| - \frac{1}{2x^2y^2} + C \tag{272}$$

□

Example 15.3 (Homogeneous Differential Form Integrating Factor)**.** Find an integrating factor for the homogeneous differential form:

$$(x^2y - 2xy^2)\, dx - (x^3 - 3x^2y)\, dy = 0 \tag{273}$$

and transform it into an exact differential.

Solution. **1. Verify Homogeneity:** $M(x, y) = x^2y - 2xy^2$ and $N(x, y) = -x^3 + 3x^2y$ are homogeneous of degree $n = 3$.

2. Apply Euler's Theorem Criterion for Homogeneous Forms: Compute the denominator $D(x, y) = xM + yN$:

$$\begin{aligned} xM + yN &= x(x^2y - 2xy^2) + y(-x^3 + 3x^2y) \\ &= x^3y - 2x^2y^2 - x^3y + 3x^2y^2 = x^2y^2 \end{aligned} \tag{274}$$

Since $xM + yN = x^2y^2 \neq 0$, the integrating factor is:

$$\mu(x, y) = \frac{1}{xM + yN} = \frac{1}{x^2y^2} \tag{275}$$

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3. Multiply and Verify Exactness:

$$\omega_\mu = \left(\frac{1}{y} - \frac{2}{x}\right) dx + \left(\frac{3}{y} - \frac{x}{y^2}\right) dy \tag{276}$$

Verify exactness of modified components:

$$\frac{\partial}{\partial y}\left(\frac{1}{y} - \frac{2}{x}\right) = -\frac{1}{y^2}, \quad \frac{\partial}{\partial x}\left(\frac{3}{y} - \frac{x}{y^2}\right) = -\frac{1}{y^2} \quad \checkmark \tag{277}$$

Reconstructing potential $f(x, y)$:

$$f(x,y) = \int \left(\frac{1}{y} - \frac{2}{x}\right) dx = \frac{x}{y} - 2\ln|x| + g(y) \tag{278}$$

$$\frac{\partial f}{\partial y} = -\frac{x}{y^2} + g'(y) = \frac{3}{y} - \frac{x}{y^2} \implies g'(y) = \frac{3}{y} \implies g(y) = 3\ln|y| \tag{279}$$

Final Result:

$$f(x,y) = \frac{x}{y} + \ln\left|\frac{y^3}{x^2}\right| + C \tag{280}$$

□

16 Exercises / Practice Problems

Exercise 16.1. Determine whether the following limit exists along different path families:

$$\lim_{(x,y)\to(0,0)} \frac{xy^3}{x^2 + y^6} \tag{281}$$

Exercise 16.2. Evaluate or show the non-existence of the limit:

$$\lim_{(x,y)\to(0,0)} \frac{x^3 y}{x^6 + y^2} \tag{282}$$

Exercise 16.3. Investigate the path-dependency of the limit:

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y^2}{x^4 + 3y^4} \tag{283}$$

Exercise 16.4. Test for limit existence by comparing paths $y = mx^4$:

$$\lim_{(x,y)\to(0,0)} \frac{x^4 y}{x^8 + y^2} \tag{284}$$

Exercise 16.5. Determine whether the following limit exists:

$$\lim_{(x,y)\to(0,0)} \frac{x^3 y^2}{x^5 + y^5} \tag{285}$$

Exercise 16.6. Evaluate the limit along parabolic paths $x = ky^2$:

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2 + y^4} \tag{286}$$

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Exercise 16.7. Determine whether the following limit exists:

$$\lim_{(x,y)\to(0,0)} \frac{2x^3y}{x^4+y^2} \tag{287}$$

Exercise 16.8. Evaluate the limit or prove non-existence:

$$\lim_{(x,y)\to(0,0)} \frac{x^2ye^y}{x^4+y^2} \tag{288}$$

Exercise 16.9. Show that the following limit does not exist using straight line paths $y = mx$:

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} \tag{289}$$

Exercise 16.10. Use polar coordinates to evaluate the limit:

$$\lim_{(x,y)\to(0,0)} \frac{x^3+y^3}{x^2+y^2} \tag{290}$$

Exercise 16.11. Evaluate the limit using the Squeeze Theorem in polar coordinates:

$$\lim_{(x,y)\to(0,0)} \frac{x^2y^2}{\sqrt{x^2+y^2}} \tag{291}$$

Exercise 16.12. Compute the limit:

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2} \tag{292}$$

Exercise 16.13. Evaluate:

$$\lim_{(x,y)\to(0,0)} \frac{1-\cos\sqrt{x^2+y^2}}{x^2+y^2} \tag{293}$$

Exercise 16.14. Prove using polar coordinates whether the limit exists:

$$\lim_{(x,y)\to(0,0)} (x^2+y^2)\ln(x^2+y^2) \tag{294}$$

Exercise 16.15. Evaluate the limit:

$$\lim_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2} \tag{295}$$

Exercise 16.16. Compute:

$$\lim_{(x,y)\to(0,0)} \frac{\tan(x^2+y^2)}{\sqrt{x^2+y^2}} \tag{296}$$

Exercise 16.17. Determine whether $f(x,y)$ is continuous at $(0,0)$:

$$f(x,y) = \begin{cases} \frac{x^2y^3}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases} \tag{297}$$

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Exercise 16.18. Determine the continuity of f at the origin:

$$f(x,y) = \begin{cases} \frac{\sin(xy)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases} \tag{298}$$

Exercise 16.19. Test $f(x,y)$ for continuity at $(0,0)$:

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases} \tag{299}$$

Exercise 16.20. Investigate continuity of the piecewise function at $(0,0)$:

$$f(x,y) = \begin{cases} \frac{x^3-3xy^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases} \tag{300}$$

Exercise 16.21. Find the value of c that makes $f(x,y)$ continuous on $\mathbb{R}^2$:

$$f(x,y) = \begin{cases} \frac{e^{x^2+y^2}-1}{x^2+y^2}, & (x,y) \neq (0,0) \\ c, & (x,y) = (0,0) \end{cases} \tag{301}$$

Exercise 16.22. Evaluate the 3D limit or show it does not exist:

$$\lim_{(x,y,z)\to(0,0,0)} \frac{xyz}{x^2+y^2+z^2} \tag{302}$$

Exercise 16.23. Test the 3D limit along $x = t, y = t, z = t$ vs. $x = t^2, y = t, z = t$:

$$\lim_{(x,y,z)\to(0,0,0)} \frac{x^2yz}{x^4+y^2+z^2} \tag{303}$$

Exercise 16.24. Evaluate using rationalization:

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2+1}-1} \tag{304}$$

Exercise 16.25. Compute the limit by shifting the origin:

$$\lim_{(x,y)\to(1,0)} \frac{(x-1)^2y}{(x-1)^2+y^2} \tag{305}$$

Exercise 16.26. Compute $f_x(0,0)$ and $f_y(0,0)$ using the formal limit definition for:

$$f(x,y) = \begin{cases} \frac{x^2y}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases} \tag{306}$$

Exercise 16.27. Compute $f_x(0,0)$ and $f_y(0,0)$ via definition for:

$$f(x,y) = \begin{cases} \frac{xy^2}{x^2+y^4}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases} \tag{307}$$

Exercise 16.28. Prove that $f_x(0,0) = 0$ and $f_y(0,0) = 0$ for $f(x,y) = \sqrt{|xy|}$.

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Exercise 16.29. Determine whether $f(x,y) = \frac{x^2 y}{x^2+y^2}$ (with $f(0,0) = 0$) is totally differentiable at $(0,0)$.

Exercise 16.30. Test total differentiability of $f(x,y) = \sqrt{|xy|}$ at $(0,0)$.

Exercise 16.31. Test total differentiability of $f(x,y) = \frac{x^3-y^3}{x^2+y^2}$ (with $f(0,0) = 0$) at the origin.

Exercise 16.32. Determine if the function $f(x,y) = (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right)$ (and $f(0,0) = 0$) is differentiable at $(0,0)$.

Exercise 16.33. Verify Clairaut's Theorem ($f_{xy} = f_{yx}$) for $f(x,y) = x^3y^2 + \sin(xy)$.

Exercise 16.34. Verify $f_{xy} = f_{yx}$ for $f(x,y) = e^{x^2 y}$.

Exercise 16.35. Verify Clairaut's Theorem for $f(x,y) = \arctan\left(\frac{y}{x}\right)$ on $x > 0$.

Exercise 16.36. Verify $f_{xy} = f_{yx}$ for $f(x,y) = x^y$ where $x > 0$.

Exercise 16.37. For the function:

$$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases} \tag{308}$$

Show that $f_{xy}(0,0) = -1$ and $f_{yx}(0,0) = 1$, thus demonstrating a failure of Clairaut's Theorem.

Exercise 16.38. Calculate the total differential df for $f(x,y) = x^2y + 3xy^3$ at the point $(1,2)$.

Exercise 16.39. Use the linear approximation of $f(x,y,z) = \sqrt{x^2+y^2+z^2}$ at $(4,3,12)$ to estimate $\sqrt{(4.05)^2 + (3.02)^2 + (12.01)^2}$.

Exercise 16.40. Use total differentials to approximate the product $(1.02)^3(0.97)^4$.

Exercise 16.41. Find the equation of the tangent plane to $z = x^2 + 3y^2 - 4x$ at $(2,1,3)$.

Exercise 16.42. Find the equation of the tangent plane to $z = e^{2x}\cos(3y)$ at $(0,0,1)$.

Exercise 16.43. Find the equation of the tangent plane to the surface $x^2 + 2y^2 + 3z^2 = 6$ at $(1,1,1)$.

Exercise 16.44. Approximate $f(2.01, 0.98)$ for $f(x,y) = x^y$ using linearization at $(2,1)$.

Exercise 16.45. Compute the second total differential d^2f for $f(x,y) = x^3 - 3xy^2 + y^3$.

Exercise 16.46. Show that $f(x,y) = |x| + |y|$ is continuous everywhere on $\mathbb{R}^2$ but fails to be differentiable at $(0,0)$.

Exercise 16.47. Show that $f(x,y) = x\sin(1/y) + y\sin(1/x)$ for $x, y \neq 0$ can be extended continuously to $(0,0)$, and test its differentiability at $(0,0)$.

Exercise 16.48. Compute the third-order mixed partial derivative f_{xyz} for $f(x,y,z) = e^{xyz}$.

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Exercise 16.49. If $f(x,y) = \ln(x^2+y^2)$, calculate the Laplacian $f_{xx}+f_{yy}$ on $\mathbb{R}^2\setminus\{(0,0)\}$.

Exercise 16.50. Linearize $f(x,y) = \sqrt{1+x+y}$ around $(0,0)$ and state the linear approximation formula.

Exercise 16.51. Let $w = x^2+y^2-z^2$, with $x = r\cos\theta$, $y = r\sin\theta$, and $z = r$. Compute $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial \theta}$ using the Chain Rule.

Exercise 16.52. Let $z = f(x,y)$ be differentiable, where $x = u^2 - v^2$ and $y = 2uv$. Express $z_u^2 + z_v^2$ in terms of z_x and z_y.

Exercise 16.53. Prove that any function of the form $u(x,t) = f(x-ct) + g(x+ct)$ satisfies the 1D wave equation $u_{tt} = c^2 u_{xx}$.

Exercise 16.54. Show that if $z = xf(y/x)$ for a differentiable function f, then $xz_x + yz_y = z$.

Exercise 16.55. Transform the partial differential equation $xu_x + yu_y = 0$ using the change of variables $\xi = x$, $\eta = y/x$.

Exercise 16.56. Given $z = f(u,v)$ with $u = x+y$ and $v = x-y$, show that $z_x z_y = z_u^2 - z_v^2$.

Exercise 16.57. Transform the wave equation $u_{tt} - c^2 u_{xx} = 0$ into characteristic coordinates $\xi = x - ct$ and $\eta = x + ct$.

Exercise 16.58. Let $u = f(r)$ where $r = \sqrt{x^2+y^2+z^2}$. Express the 3D Laplacian $u_{xx} + u_{yy} + u_{zz}$ purely in terms of r and derivatives of f.

Exercise 16.59. Let $z = e^x \cos y$, where $x = t^3 + 1$ and $y = \sqrt{t}$. Compute $\frac{dz}{dt}$ at $t = 1$.

Exercise 16.60. Let $w = xy + yz + zx$, where $x = st$, $y = s/t$, and $z = s + t$. Compute $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$.

Exercise 16.61. Show that $z = f(x^2+y^2)$ satisfies the partial differential equation $yz_x - xz_y = 0$.

Exercise 16.62. If $z = f(x+ay) + g(x-ay)$, prove that $z_{yy} = a^2 z_{xx}$.

Exercise 16.63. Transform $u_{xx} - u_{yy} = 0$ using the rotation coordinates $\xi = x + y$ and $\eta = x - y$.

Exercise 16.64. Transform the 3D Laplace equation $u_{xx} + u_{yy} + u_{zz} = 0$ into spherical coordinates (r, θ, ϕ).

Exercise 16.65. Let $z = f(x,y)$ where $x = e^u \cos v$ and $y = e^u \sin v$. Show that:

$$z_{xx} + z_{yy} = e^{-2u}(z_{uu} + z_{vv}) \tag{309}$$

Exercise 16.66. Find $\frac{d}{dt} f(x(t), y(t), z(t))$ for $f(x,y,z) = xyz$, $x = \sin t$, $y = \cos t$, $z = e^t$.

Exercise 16.67. Compute $\frac{\partial^2 z}{\partial u \partial v}$ if $z = f(x,y)$, $x = uv$, and $y = u/v$.

Exercise 16.68. Show that any function $w = f(x-y, y-z, z-x)$ satisfies $w_x + w_y + w_z = 0$.

Exercise 16.69. Convert the operator $x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}$ into polar coordinates (r, θ).

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Exercise 16.70. If $u = f(x^2 - y^2, y^2 - z^2)$, show that $\frac{1}{x}u_x + \frac{1}{y}u_y + \frac{1}{z}u_z = 0$.

Exercise 16.71. Compute $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ for $z = x^2y - y^3x$ under polar coordinate substitutions.

Exercise 16.72. Show that $z = \frac{1}{y}f(y/x)$ satisfies $xz_x + yz_y + z = 0$.

Exercise 16.73. State and prove Euler's Theorem for homogeneous functions of degree n in two variables using the multivariable chain rule.

Exercise 16.74. Formulate the second-order chain rule expansion for $\frac{d^2}{dt^2}f(x(t), y(t))$.

Exercise 16.75. Transform the second-order PDE $2u_{xx} + 5u_{xy} + 2u_{yy} = 0$ into canonical form using an appropriate linear change of variables.

Exercise 16.76. Given $x^3 + y^3 + z^3 - 3xyz = 0$, compute $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Exercise 16.77. Given $\sin(x + y) + \cos(y + z) = 1$, evaluate $\frac{\partial z}{\partial x}$ at $(0, \pi/2, 0)$.

Exercise 16.78. Given $xy + yz + zx = 1$, compute $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Exercise 16.79. Given $xe^y + ye^z + ze^x = 0$, compute $\frac{\partial z}{\partial x}$ at $(0, 0, 0)$.

Exercise 16.80. Find $\frac{dy}{dx}$ for the implicitly defined relation $x^y = y^x$ where $x, y > 0$.

Exercise 16.81. Compute $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = r^2$ via implicit differentiation.

Exercise 16.82. Compute $\frac{d^2y}{dx^2}$ at (a, a) for the Folium of Descartes $x^3 + y^3 = 3axy$.

Exercise 16.83. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for the cubic relation $z^3 - xz - y = 0$.

Exercise 16.84. Find $\frac{\partial z}{\partial x}$ for $\ln(x + z) + xyz = 1$.

Exercise 16.85. Find the second partial derivative $\frac{\partial^2 z}{\partial x^2}$ for the sphere equation $x^2 + y^2 + z^2 = R^2$.

Exercise 16.86. Consider the implicit system $u + v = x + y$ and $uv = x - y$. Compute $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial y}$.

Exercise 16.87. For the polar system $x = u\cos v, y = u\sin v$, compute the Jacobian matrix $J_{(u,v)}(x, y)$ and its inverse matrix $J_{(x,y)}(u, v)$.

Exercise 16.88. Given the system $u^2 + v^2 = x$ and $u - v = y$, compute $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$.

Exercise 16.89. Consider $F(x, y, u, v) = xu + yv - 1 = 0$ and $G(x, y, u, v) = u^2 + v^2 - x^2 - y^2 = 0$. Compute $\frac{\partial u}{\partial x}$ using Jacobians.

Exercise 16.90. Given $u^3 + v^3 = x + y$ and $u^2 + v^2 = x - y$, compute $\frac{\partial u}{\partial x}$ using Cramer's Rule.

Exercise 16.91. Given $x + y + z + u = 0$ and $x^2 + y^2 + z^2 + u^2 = 1$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial u}{\partial x}$ treating z, u as dependent variables.

Exercise 16.92. Verify the Implicit Function Theorem hypotheses for $F(x, y, z) = x^2 + y^2 + z^2 - 14 = 0$ near $(1, 2, 3)$, and write the tangent plane equation to the surface at this point.

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Exercise 16.93. For $u = x^2 - y^2$ and $v = 2xy$, prove local invertibility everywhere on $\mathbb{R}^2 \setminus \{(0,0)\}$.

Exercise 16.94. For the system $x = u + v + w$, $y = u^2 + v^2 + w^2$, $z = u^3 + v^3 + w^3$, compute the Jacobian determinant $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.

Exercise 16.95. Compute $\frac{\partial z}{\partial x}$ for the surface defined implicitly by $f(x/z, y/z) = 0$.

Exercise 16.96. Prove the triple product rule for cyclic implicit variables:

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1 \tag{310}$$

Exercise 16.97. Given $e^u + e^v = x$ and $e^{-u} + e^{-v} = y$, compute $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial y}$.

Exercise 16.98. Find $\frac{dy}{dx}$ if y is defined implicitly by $\int_0^{x+y} \sin(t^2)\,dt = x$.

Exercise 16.99. Compute $\frac{\partial z}{\partial x}$ at $(1,1,1)$ for $x^2yz + y^2zx + z^2xy = 3$.

Exercise 16.100. For $F(u,v) = 0$ where $u = x^2 + y^2$ and $v = xy$, find $\frac{dy}{dx}$ in terms of x, y, F_u, F_v.

Exercise 16.101. Test exactness of $\omega = (2xy + \cos y)\,dx + (x^2 - x\sin y + 3y^2)\,dy$ and find its potential function $f(x,y)$.

Exercise 16.102. Test exactness of $\omega = (ye^{xy} + 4x^3)\,dx + (xe^{xy} - 2y)\,dy$ and reconstruct $f(x,y)$.

Exercise 16.103. Determine whether $\omega = \left(3x^2y^2 + \frac{1}{x}\right)dx + (2x^3y + \cos y)dy$ is exact, and if so, reconstruct $f(x,y)$.

Exercise 16.104. Reconstruct the potential function for $df = (y^2\cos x + 2xy)\,dx + (2y\sin x + x^2 - 4)\,dy$.

Exercise 16.105. Reconstruct $f(x,y)$ if $df = \frac{y\,dx - x\,dy}{x^2+y^2}$. Discuss the exactness and domain restriction of this form.

Exercise 16.106. Reconstruct the 3D potential function $f(x,y,z)$ for:

$$df = (2xyz + z^2)\,dx + (x^2z + 3y^2)\,dy + (x^2y + 2xz + 1)\,dz \tag{311}$$

Exercise 16.107. Reconstruct the 3D potential function $f(x,y,z)$ for:

$$df = (yze^{xyz} + 2x)\,dx + (xze^{xyz} - 3y^2)\,dy + (xye^{xyz} + 4z^3)\,dz \tag{312}$$

Exercise 16.108. Reconstruct the 3D potential function $f(x,y,z)$ for:

$$df = \left(\frac{1}{y} + z\right)dx + \left(-\frac{x}{y^2} + 2yz\right)dy + (x + y^2)dz \tag{313}$$

Exercise 16.109. Use path-independent definite line integration from $(0,0)$ to (x,y) to reconstruct $f(x,y)$ for $df = (6xy + 2y^2)\,dx + (3x^2 + 4xy - 5)\,dy$ given $f(0,0) = 2$.

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Exercise 16.110. Use path integration from $(1,1)$ to (x,y) to find $f(x,y)$ for $df = \frac{x\,dx+y\,dy}{x^2+y^2}$ with $f(1,1)=0$.

Exercise 16.111. Find an integrating factor $\mu(x)$ depending only on x for the differential equation:

$$(3x^2y + 2xy + y^3)\,dx + (x^2 + y^2)\,dy = 0 \tag{314}$$

Exercise 16.112. Find an integrating factor $\mu(y)$ depending only on y for:

$$(2xy^4e^y + 2xy^3 + y)\,dx + (x^2y^4e^y - x^2y^2 - 3x)\,dy = 0 \tag{315}$$

Exercise 16.113. Find an integrating factor $\mu(x)$ for $(x^2 + y^2 + x)\,dx + xy\,dy = 0$.

Exercise 16.114. Find an integrating factor $\mu(y)$ for $y(2x + y^2)\,dx + x(y^2 - x)\,dy = 0$.

Exercise 16.115. Find an integrating factor of the form $\mu = \mu(z)$ where $z = xy$ for:

$$(y + x^3y^4)\,dx + (x + x^4y^3)\,dy = 0 \tag{316}$$

Exercise 16.116. Find an integrating factor of the form $\mu = \mu(z)$ where $z = xy$ for:

$$(2y + 3xy^2)\,dx + (x + 2x^2y)\,dy = 0 \tag{317}$$

Exercise 16.117. Find an integrating factor of the form $\mu = \mu(x + y)$ for:

$$(3x + 2y + y^2)\,dx + (x + 4y + xy)\,dy = 0 \tag{318}$$

Exercise 16.118. Find the integrating factor $\mu(x,y) = \frac{1}{xM+yN}$ for the homogeneous form $(x^3 + y^3)\,dx - xy^2\,dy = 0$.

Exercise 16.119. Find an integrating factor for the homogeneous form $(y^2 - 2xy)\,dx + (x^2 - 2xy)\,dy = 0$ using Euler's theorem.

Exercise 16.120. Find an integrating factor for the homogeneous differential equation $(x^2y + y^3)\,dx - x^3\,dy = 0$.

Exercise 16.121. Solve $(x^2 + y^2 + 2x)\,dx + 2y\,dy = 0$ by determining an appropriate integrating factor.

Exercise 16.122. Find the necessary and sufficient conditions on the real constants a, b, c, d such that $(ax + by)\,dx + (cx + dy)\,dy = 0$ is an exact differential form.

Exercise 16.123. If $\omega = M\,dx + N\,dy$ possesses an integrating factor $\mu(x,y)$, state the precise partial differential equation satisfied by $\mu(x,y)$.

Exercise 16.124. Reconstruct $f(x,y,z)$ for the exact differential form:

$$df = \frac{x\,dx + y\,dy + z\,dz}{\sqrt{x^2 + y^2 + z^2}} \tag{319}$$

Exercise 16.125. Determine whether $\omega = (y^2z^3 + 2x)\,dx + (2xyz^3 - 3y^2)\,dy + (3xy^2z^2 + 1)\,dz$ is exact, and if so, reconstruct its scalar potential function $f(x,y,z)$.

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